Introduction to Cancer

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Quote / History

"There is no treatment.“

1600 B.C. Egypt

(comment on breast cancer)

Ancient Awareness

- 80 Million years ago – Dinosaur bones show evidence of cancer
- 3000 BC - Egyptian mummies – bone cancer
- 1600 BC – Egypt – 8 cases of breast tumors (or ulcers)
  Treated by cauterization – with the "fire drill" - Edwin Smith Papyrus

Ancient Awareness

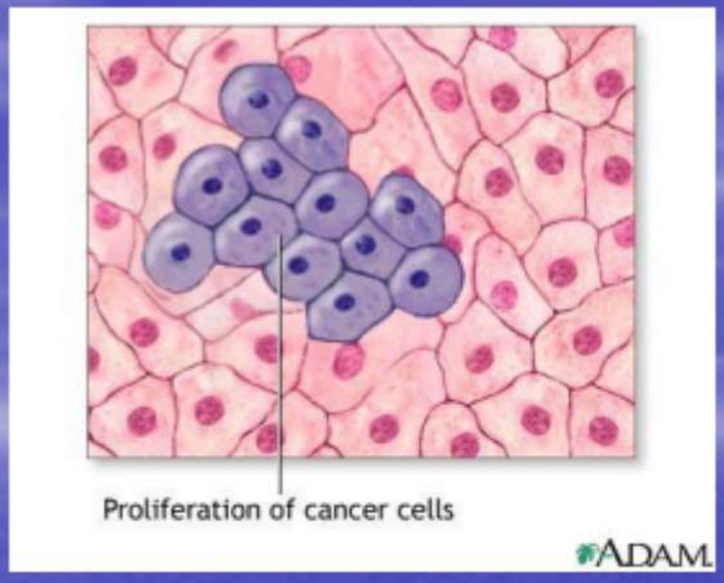
- 300 BC – Hippocrates named tumors as carcinos or carcinoma – tumors spread out like legs of a crab
- 1500 – autopsy start to provide a greater understanding of cancer
- 1650 – more knowledge with advance in medical science tools like the microscope

Human Cancer Awareness

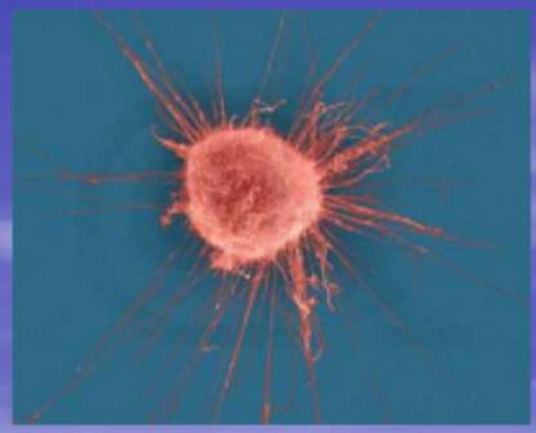
1700 – Occupational cancer – High incidence of breast cancer among nuns

1775 – Percivall Pott – Occupational cancer of scrotum in chimney sweeps

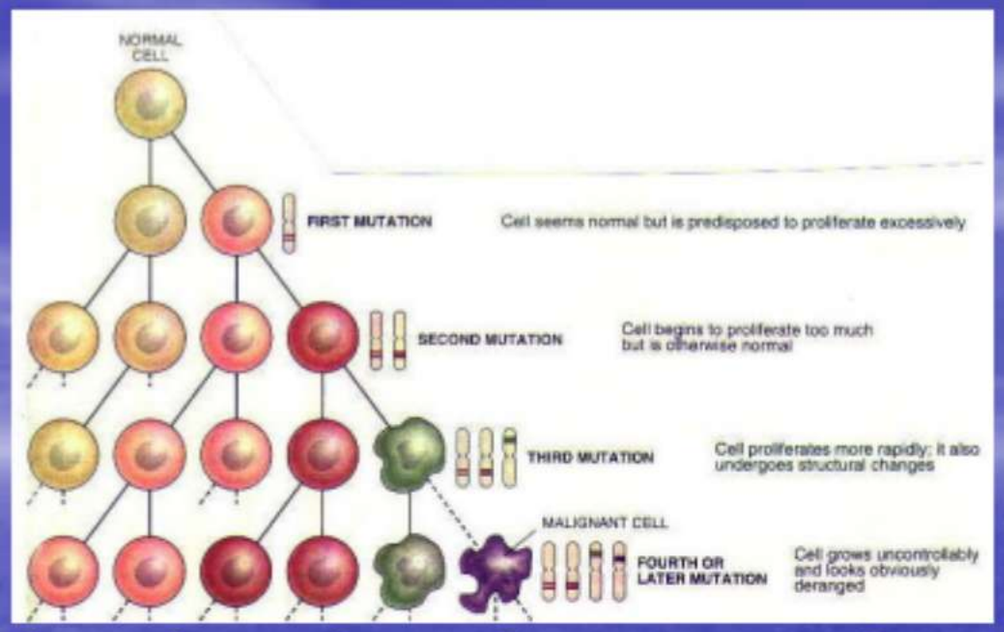
1895 – Bladder cancer in workers in aniline dye industry



- What is Cancer?
  - Normal body cells grow, divide, and die in an orderly fashion.
  - During the early years of a person's life, normal cells divide more rapidly until the person becomes an adult.
  - After that, cells in most parts of the body divide only to replace worn-out or dying cells and to repair injuries.
- Cancer develops when cells in a part of the body begin to grow out of control. Although there are many kinds of cancer, they all start because of out-of-control growth of abnormal cells.



- What is Cancer?
  - Because cancer cells continue to grow and divide, they are different from normal cells. Instead of dying, they outlive normal cells and continue to form new abnormal cells.
  - Cancer cells develop because of damage to DNA. This substance is in every cell and directs all its activities. Most of the time when DNA becomes damaged the body is able to repair it. In cancer cells, the damaged DNA is not repaired. People can inherit damaged DNA, which accounts for inherited cancers. Many times though, a person's DNA becomes damaged by exposure to something in the environment, like smoking.
    - Source - ACS



- What is the molecular basis of cancer?
  - Cancer is a genetic disease.
    - Mutations in genes result in altered proteins
      - During cell division
      - External agents
      - Random event
    - Most cancers result from mutations in somatic cells
    - Some cancers are caused by mutations in germline cells



- Theories of cancer genesis
  - Standard Dogma
    - Proto-oncogenes (Ras – melanoma)
    - Tumor suppressor genes (p53 – various cancers)
  - Modified Dogma
    - Mutation in a DNA repair gene leads to the accumulation of unrepaired mutations (Loeb, 1974) (xeroderma pigmentosum)
  - Early-Instability Theory
    - Master genes required for adequate cell reproduction are disabled, resulting in aneuploidy (Philadelphia chromosome)



- Problem posing
  - Can we understand the mechanism(s) of cancer by examining the expression patterns of genes in the cancer cell?
  - Can we use gene expression patterns to determine the properties of a cancer?

Homework

- Identify the **3** most prevalent cancers for male/female in Malaysia
- Identify the **3** most prevalent cancers for male/female in Sarawak

Learning Objectives

- The students will be able to:
  - Define cancer
  - Explain why cancer is a genetic disease even though its heritability is rather low
  - Compare the functions of oncogenes and tumor suppressor genes
  - Explain why tumor suppressors are often the 1st genes mutated in a developing cancer
  - List the Six Hallmarks of Cancer and the addition
  - Describe the advantages of using modern, targeted cancer therapies

What is Cancer?

- Cancer is the uncontrolled multiplication of cells.
- Benign – cancerous cells are contained in one place
- Malignant – cancerous have spread to other areas

What is cancer?

- Abnormal cell growth (**neoplasia**)
- **Malignant** as opposed to benign
  - **Benign:** slow growth, non-invasive, no metastasis
  - **Malignant:** rapid growth, **invasive**, potential for metastasis

What is Cancer?

- Simplest definition

  From the American Cancer Society

  " cancer is a group of diseases characterized by uncontrolled growth and spread of abnormal cells. If the spread is not controlled, it can result in death."

- Tumor
  - Two types:
    - ***Benign*** (non-cancerous) – this is *not* cancer!
      - Does not spread; it can eventually become malignant in some cases.
    - ***Malignant*** (cancerous) – this is cancer!
      - Has the potential to spread to other parts of body.

What causes cancer?

- Organic chemicals
- Inorganic chemicals
- Hormonal Carcinogenesis
- Mixtures

What causes cancer?

- **Internal factors**
  - hormones
  - immune conditions
  - inherited conditions
- **External factors**
  - lifestyle habits (smoking, diet, alcohol)
  - viruses, chemicals, radiation

Is cancer a heritable disease?

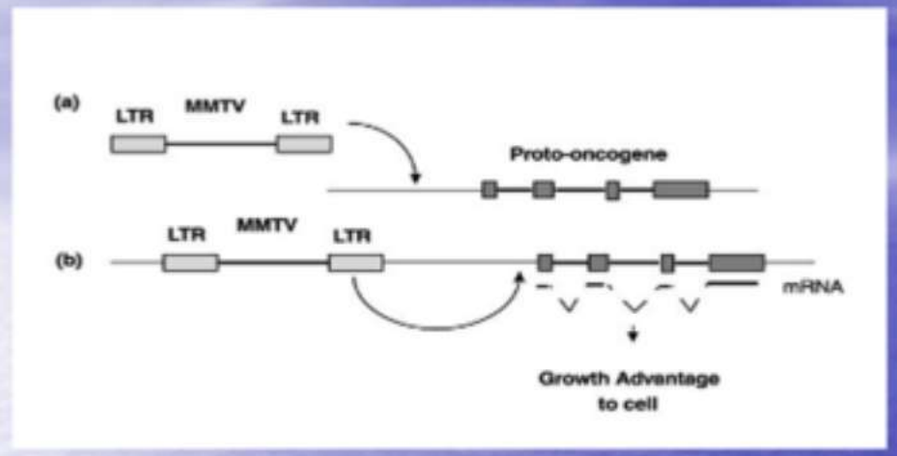
- There are heritable cancer syndromes
- The majority of cancers, however, are not familial
- Cancer is a genetic disease, but the majority of mutations that lead to cancer are **somatic**

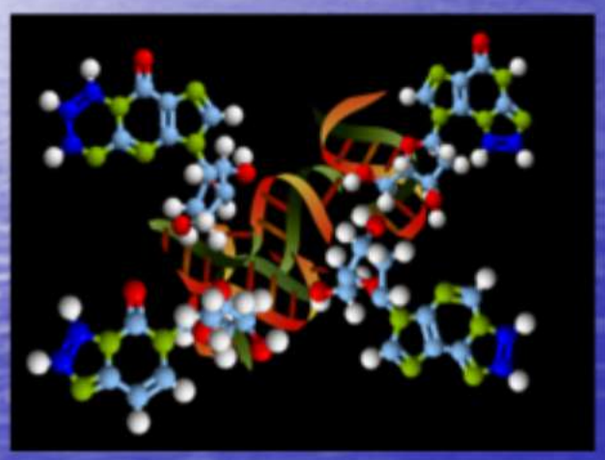
What causes the mutations that lead to cancer?

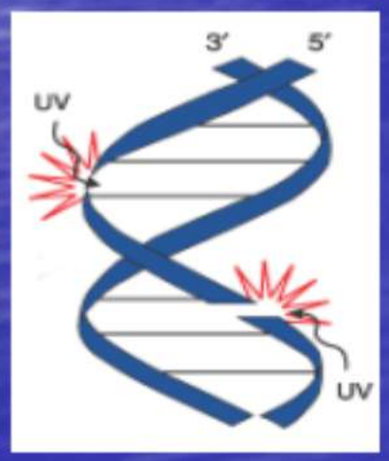
- Viruses: HPV --> cervical cancer
- Bacteria: H. pylori --> gastric cancer
- Chemicals --> B[a]P --> lung cancer
- UV and ionizing radiation --> skin cancer

- What do these agents have in common?

Mutagens







- Viruses: insertional mutagenesis
- Chemicals: DNA adducts
- UV and ionizing radiation: single and double strand DNA breaks

Cancer is a genetic disease

- Cancer arises from the *accumulation* of genetic changes (mutations).
- Most cancers have a minimum of 6-9 different genes mutated.
- Not a hereditary disease – we do not pass on cancer to offspring.
- We can inherit dispositions (susceptibility) to cancer.
- Many genes that are mutated in cancer are involved in *regulating the cell cycle.*

What types of genes get mutated in cancer?

- **Oncogenes** are activated
  - Normal function: cell growth, gene transcription
- **Tumour suppressor** genes are inactivated
  - Normal function: DNA repair, cell cycle control, cell death

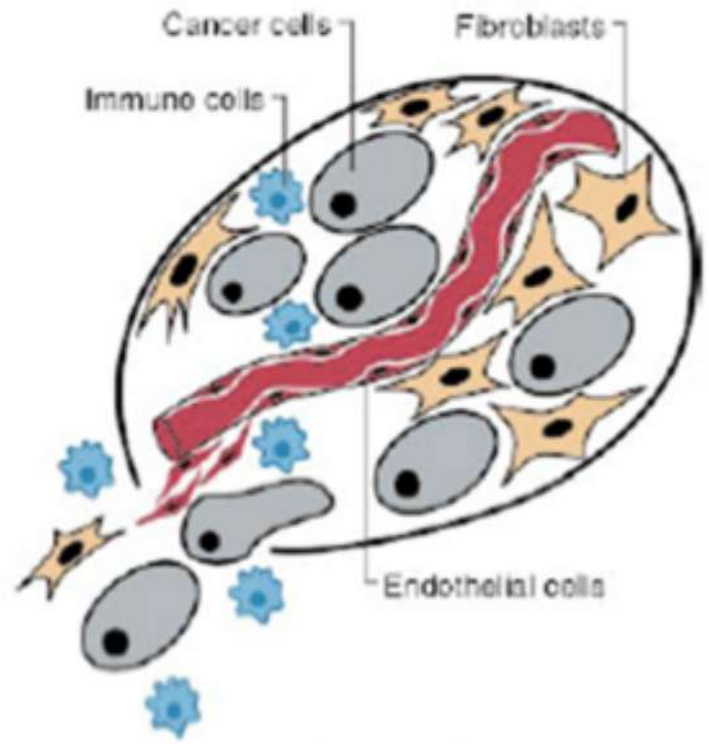
Phenotype of a cancer cell

- **The Six Hallmarks of Cancer**
  - Self-sufficient growth signals
    - Constitutively **activated** growth factor signalling
  - Resistance to anti-growth signals
    - **Inactivated** cell cycle checkpoint
  - Immortality
    - **Inactivated** cell death pathway

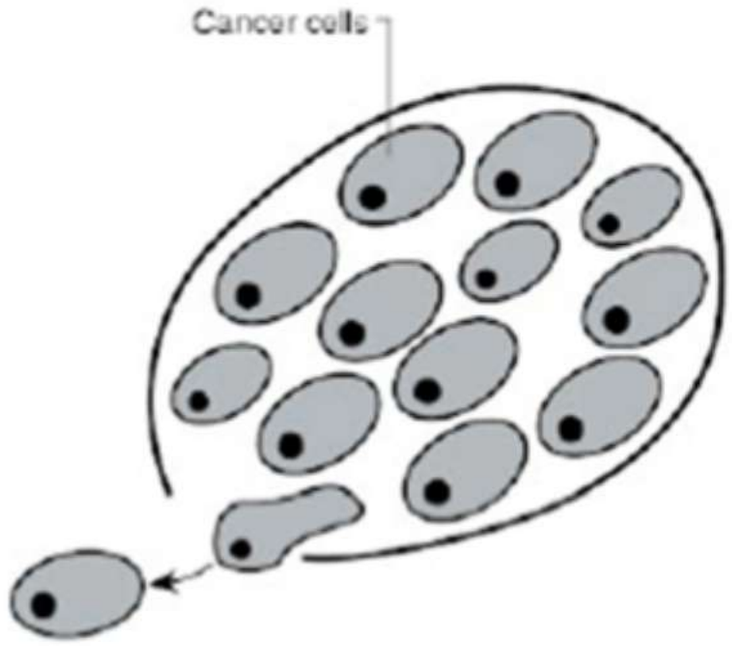
Phenotype of a cancer cell (cont'd)

- **The Six Hallmarks of Cancer**
  - Resistance to cell death
    - **Activated** anti- cell death signalling
  - Sustained angiogenesis
    - **Activated** VEGF signalling
  - Invasion and metastasis
    - Loss of cell-to-cell interactions, etc.

A Heterotypic Cell Biology



The Reductionist View



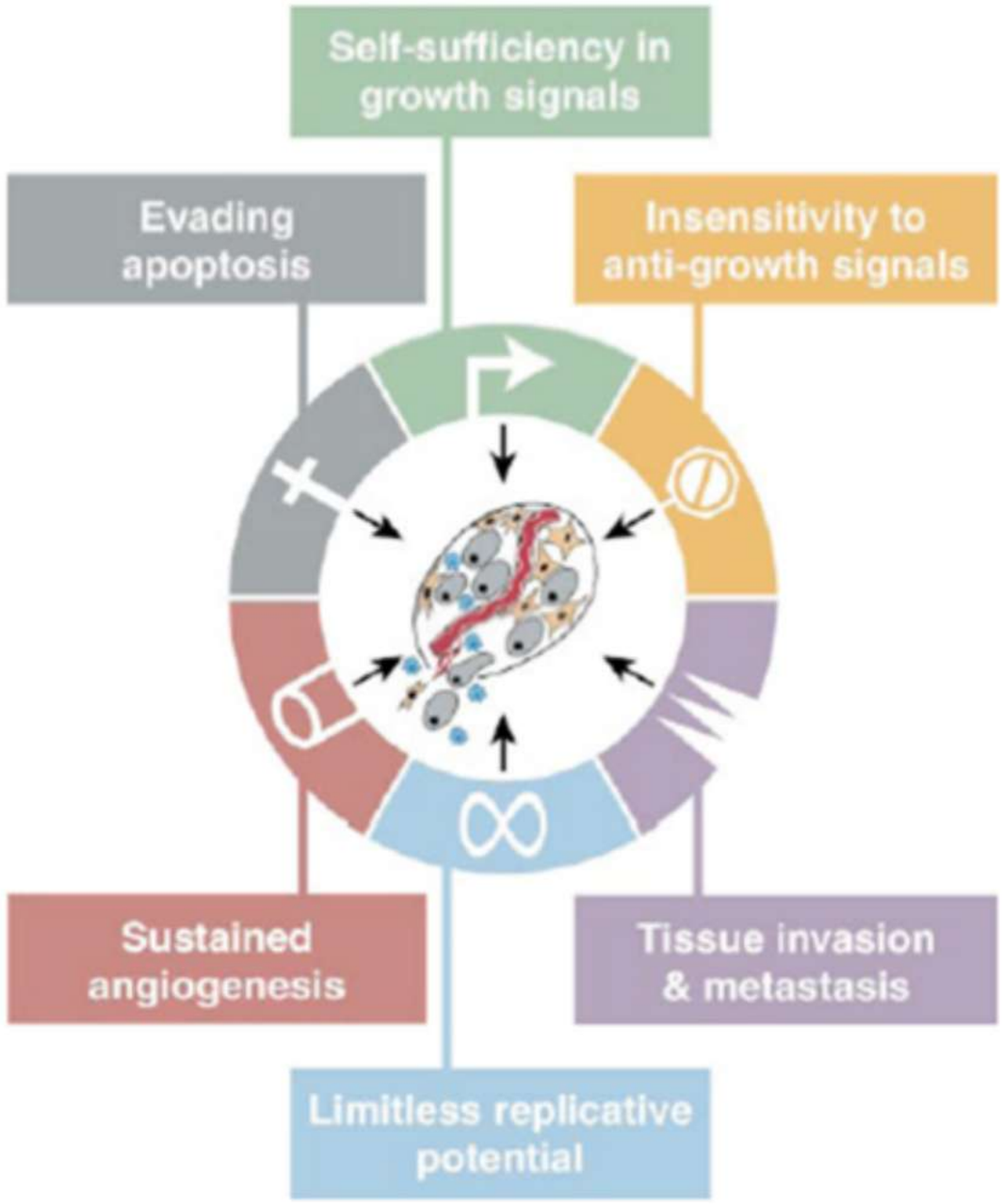
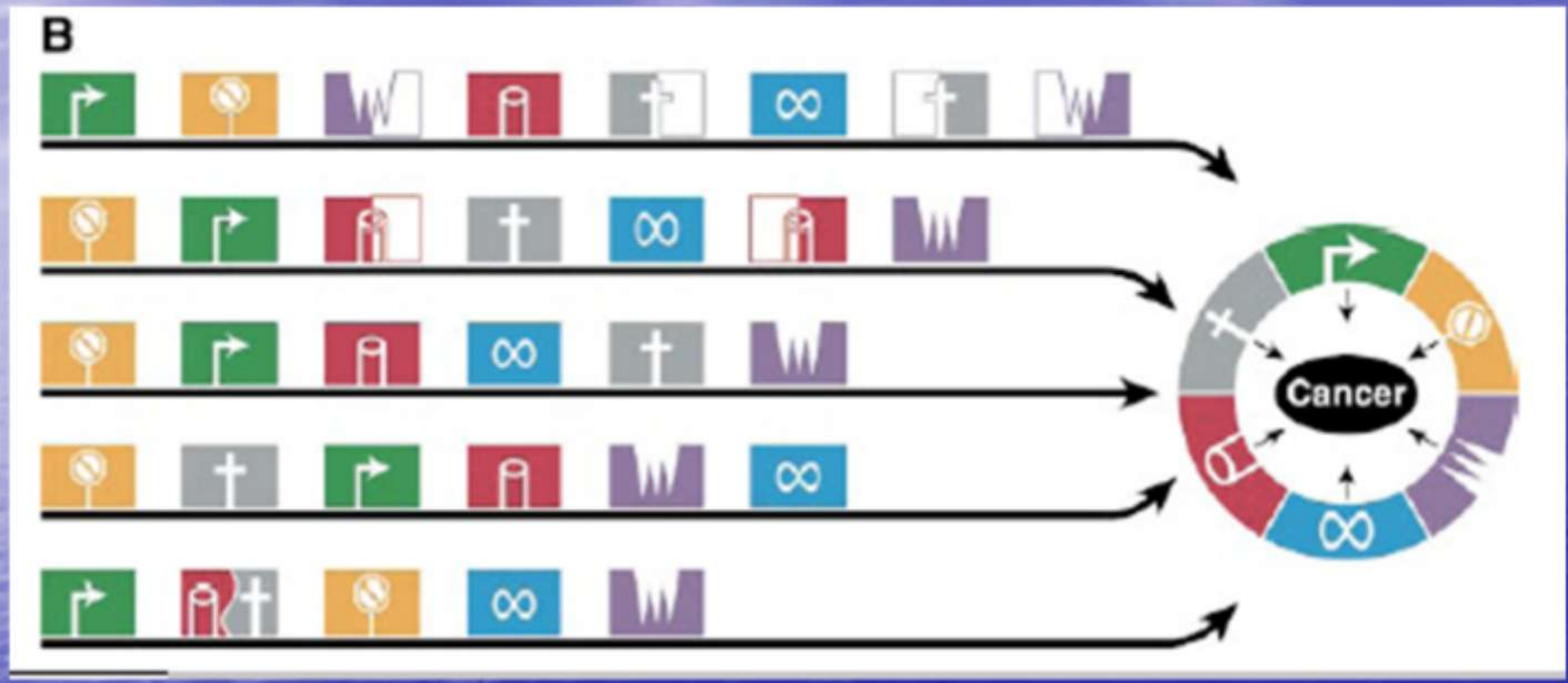


Figure 1. Acquired Capabilities of Cancer

We suggest that most if not all cancers have acquired the same set of functional capabilities during their development, albeit through various mechanistic strategies.

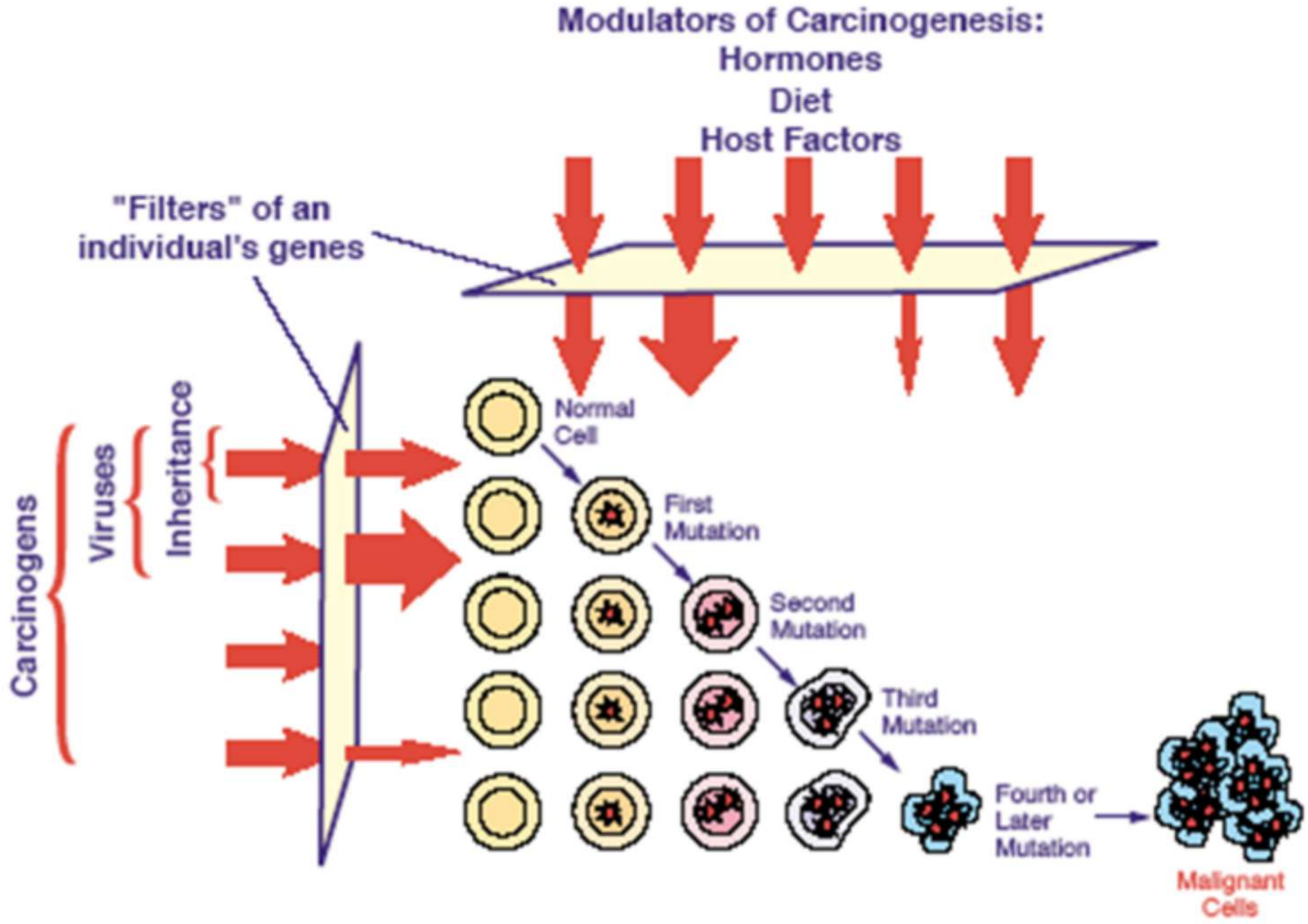
A

Component	Acquired Capability	Example of Mechanism
	Self-sufficiency in growth signals	Activate H-Ras oncogene
	Insensitivity to anti-growth signals	Lose retinoblastoma suppressor
	Evading apoptosis	Produce IGF survival factors
	Limitless replicative potential	Turn on telomerase
	Sustained angiogenesis	Produce VEGF inducer
	Tissue invasion & metastasis	Inactivate E-cadherin


B
Cancer

HOMEWORK
Hallmarks-the next generation

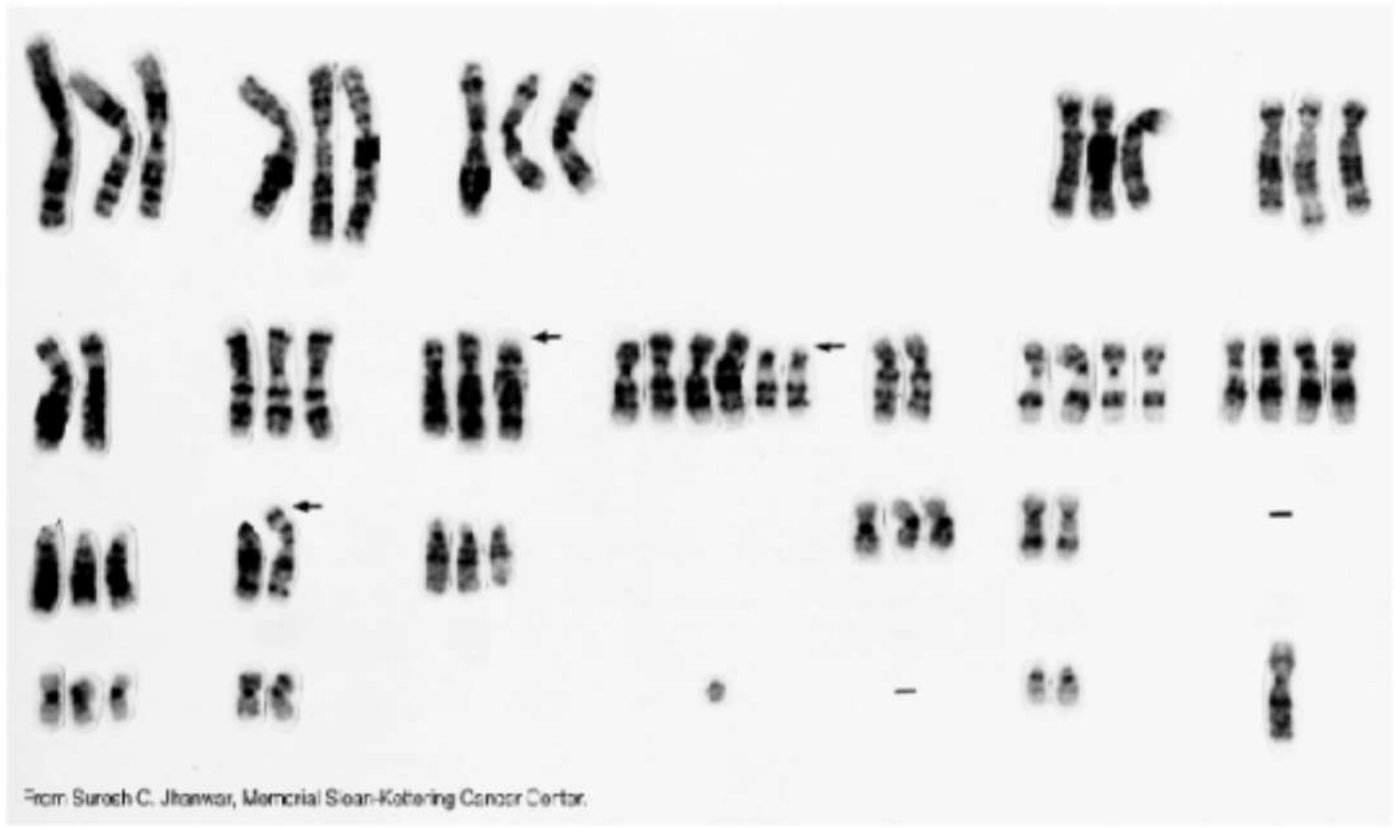
Find out what are the other aspects that are emerging as hallmarks of cancer.


Modulators of Carcinogenesis:
Hormones
Diet
Host Factors
"Filters" of an individual's genes
Carcinogens
Viruses
Inheritance
Normal Cell
First Mutation
Second Mutation
Third Mutation
Fourth or Later Mutation
Malignant Cells

Chromosomal Instability

Karyotype of a tumor cell

A highly abnormal complement of chromosomes (60 chromosomes instead of normal 46 chromosomes)



From Suresh C. Jhanwar, Memorial Sloan-Kettering Cancer Center.

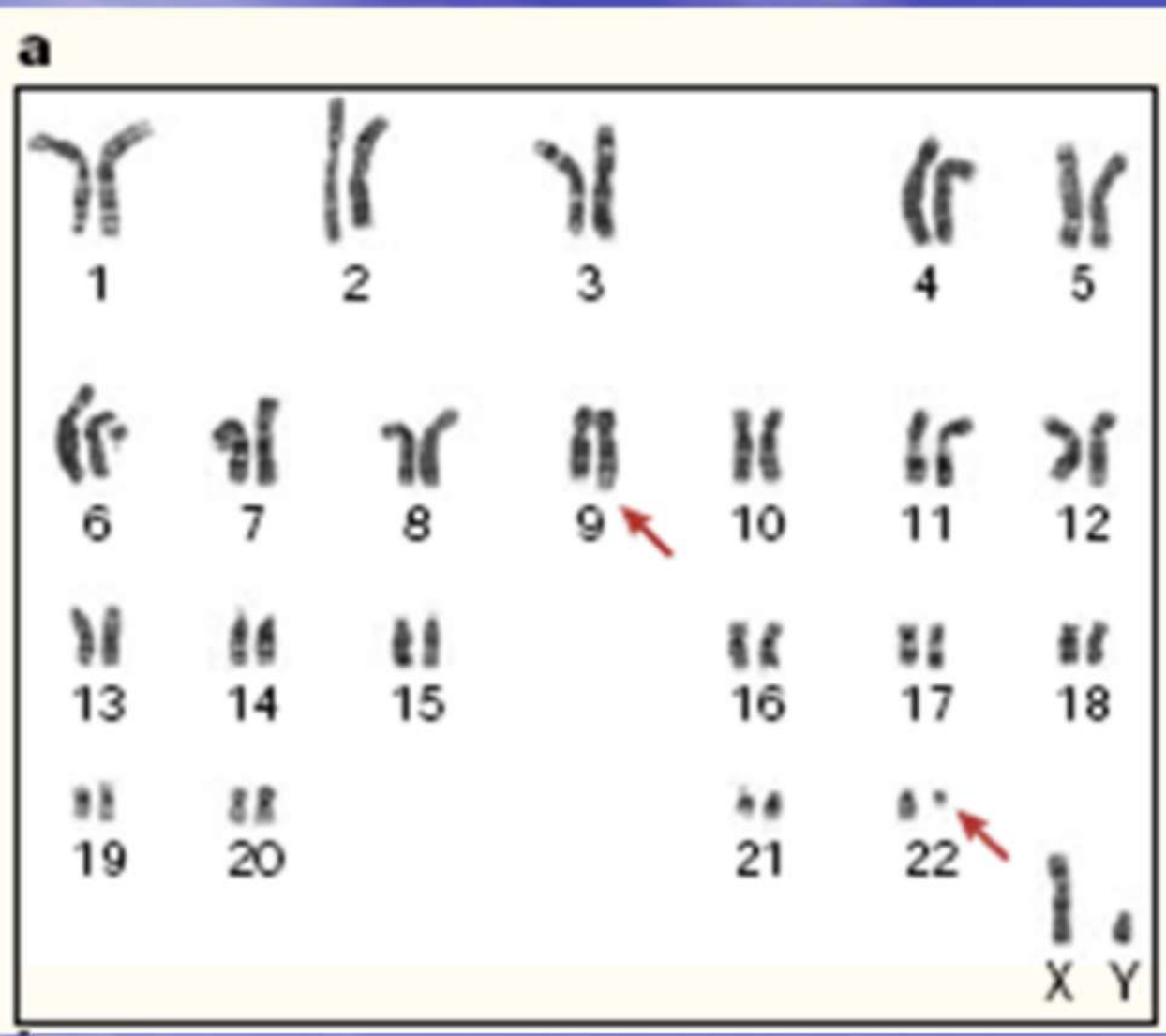


Figure 2 | **A comparison of karyotypes. a** | Chronic myelogenous leukaemia, showing the typical 9;22 translocation and an otherwise normal karyotype. **b** | Non-small-cell carcinoma of the lung, showing abnormalities of both number and structure. The arrows indicate aberrant chromosomes.

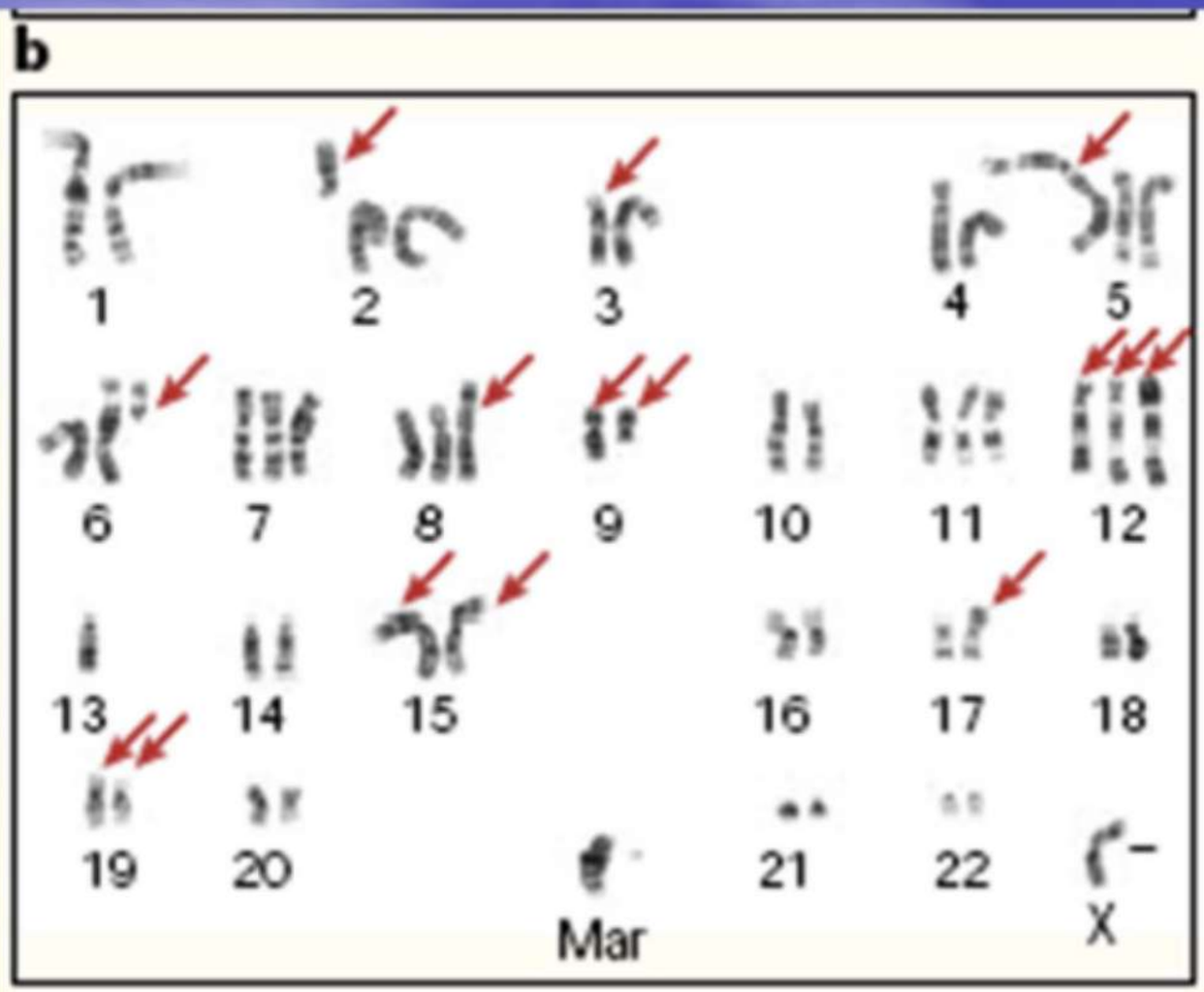


Figure 2 | **A comparison of karyotypes. a** | Chronic myelogenous leukaemia, showing the typical 9;22 translocation and an otherwise normal karyotype. **b** | Non-small-cell carcinoma of the lung, showing abnormalities of both number and structure. The arrows indicate aberrant chromosomes.

Incidence of Human Cancer

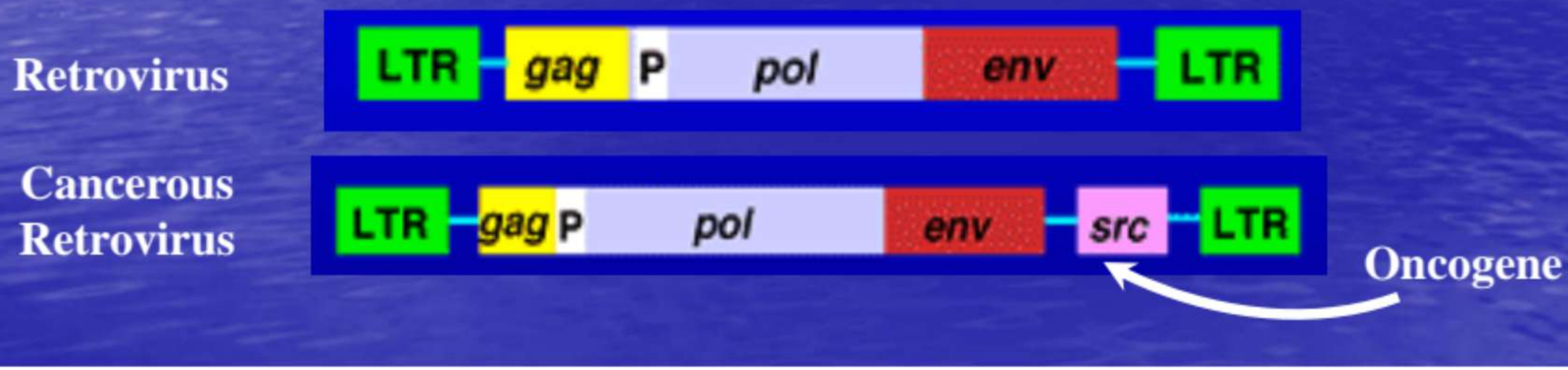
- benign tumor
- malignant tumor
- metastases
- divided into three groups
  - carcinomas
  - sarcomas
  - leukemias and lymphomas
- carcinogens
- viruses
- single mutation

Tumor suppressors and oncogenes

- Mutations in *oncogenes* and *tumor suppressor genes* can lead to cancer.

Retroviruses

- Retroviruses are members of one family of RNA viruses that cause cancer in variety of animals and humans.
- The Retrovirus is made of 3 main genes *gag, pol & env* that are required for *virus replication* but not play role in *cell transformation.*
- a retrovirus can transform cells from normal to cancer if they include a specific gene that is capable of inducing cell transformation this gene is known as <u>"Oncogene".</u>

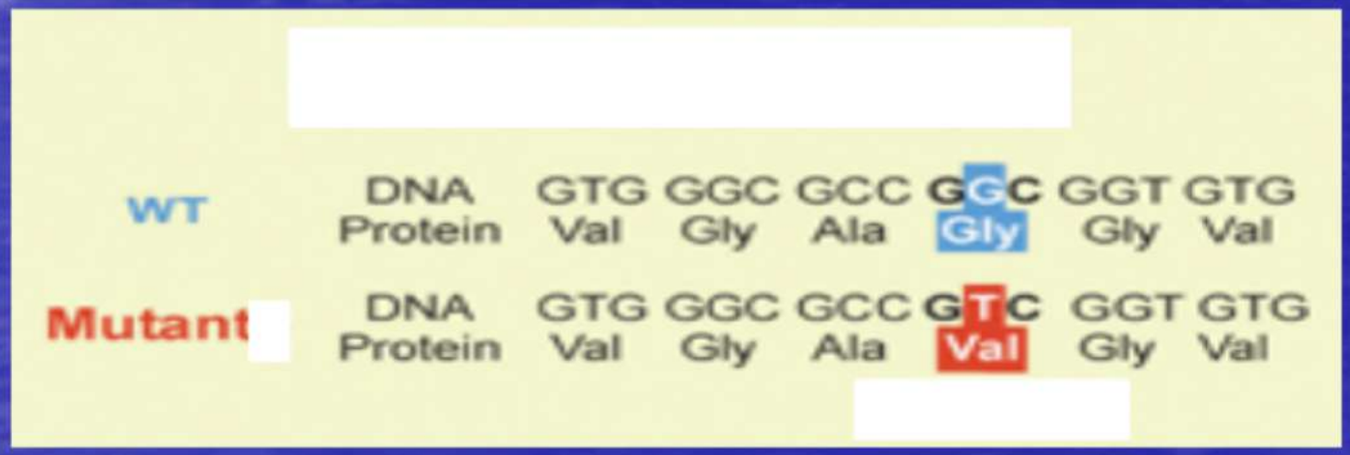


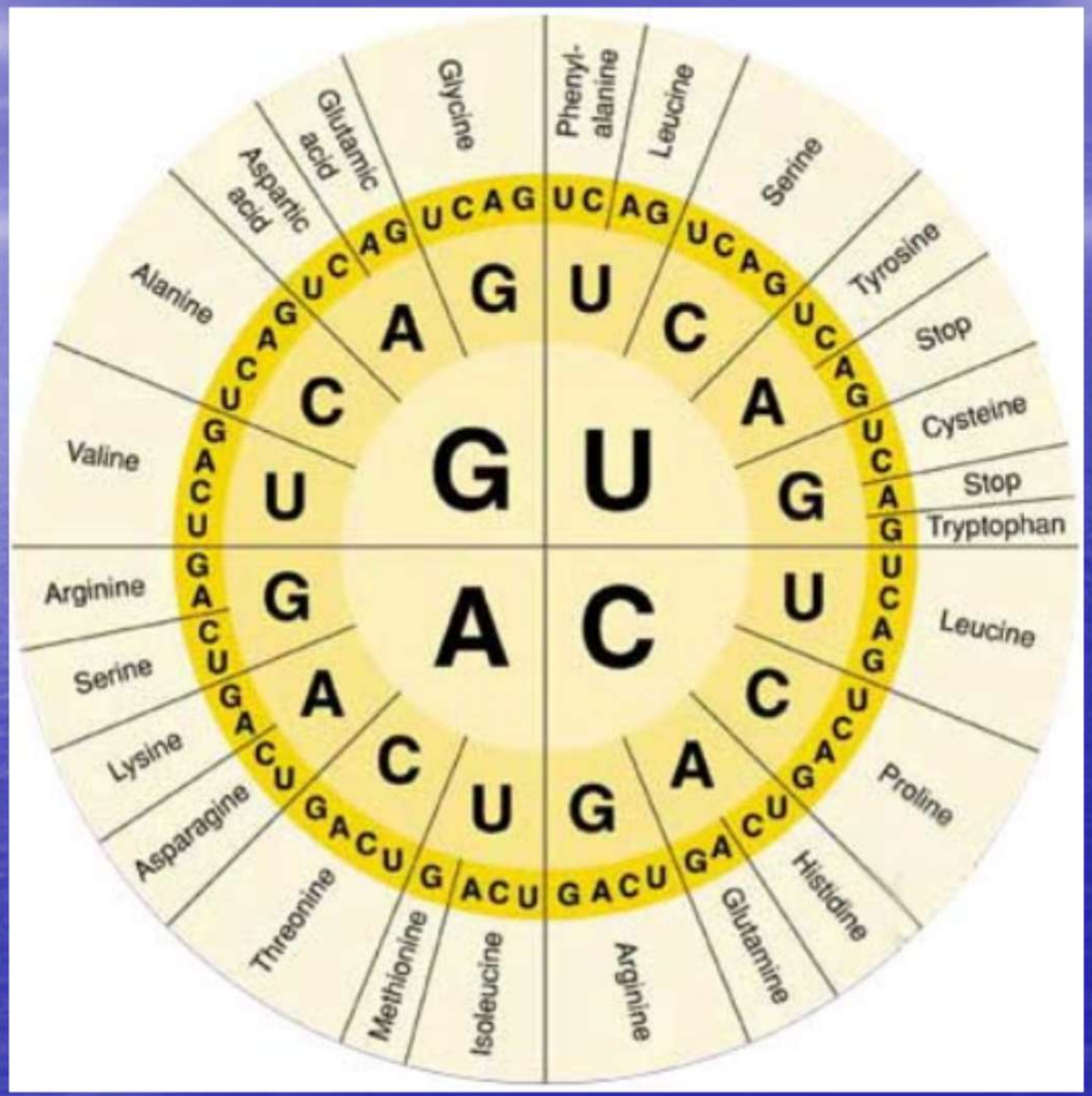
Retrovirus oncogene

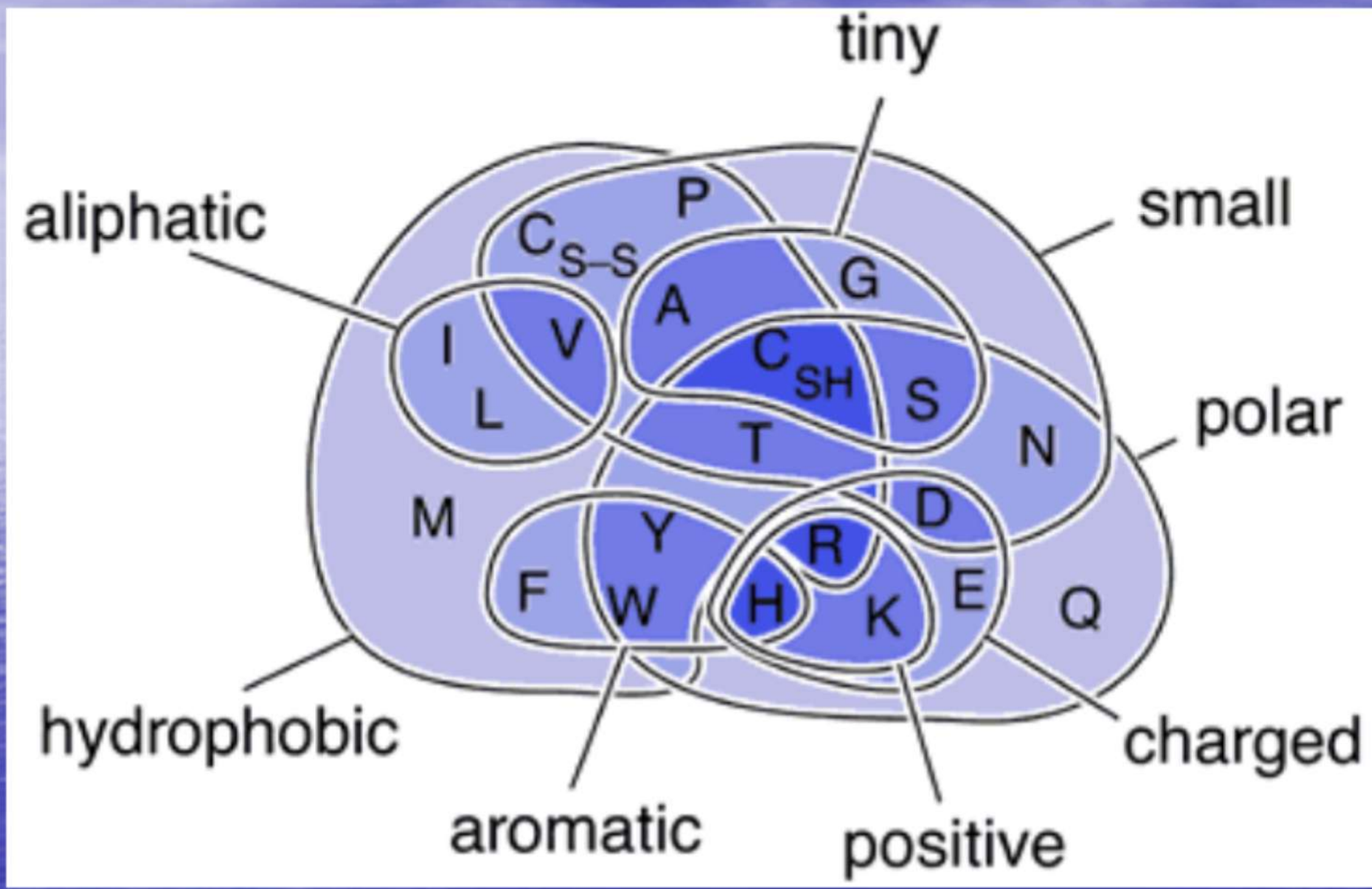
- Two main types of oncogenes:
  - *Viral oncogene*: gene from the retrovirus itself
  - *Non-Viral oncogene (Cellular oncogene):* genes derived from the genes of the host cell that are in an inactive form usually. Occasionally if the gene incorporates with the viral genome will form a highly oncogenic virus.
- Proto-oncogenes: are the form of cellular genes that inactive normally but can incorporate with the viral genome to produce a highly oncogenic virus.

Proto-Oncogene ⟶ Oncogene

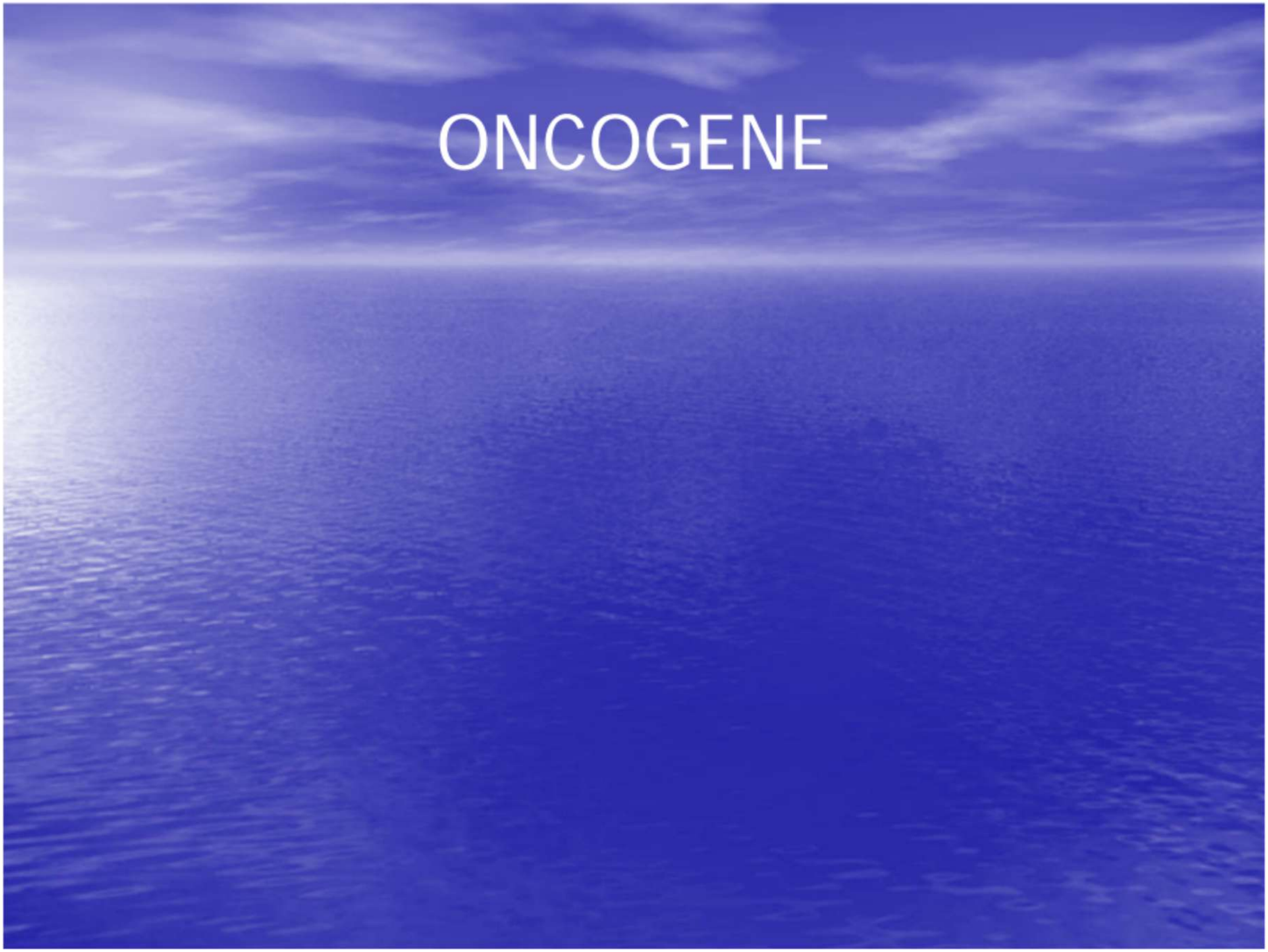
- The proto-oncogene become oncogene by:
  1. Mutation:
     - Example: mutation in *Ras gene* ⟶ Continuous activation of Ras by (constitutively in the GTP-bound conformation) ⟶ Unregulated cell proliferation Cell transformation.




U
C
A
G
Phenyl-alanine
Leucine
Serine
Tyrosine
Stop
Cysteine
Stop
Tryptophan
Leucine
Proline
Histidine
Glutamine
Arginine
Isoleucine
Methionine
Threonine
Asparagine
Lysine
Serine
Arginine
Valine
Alanine
Aspartic acid
Glutamic acid
Glycine



sulfur or thiol


ONCOGENE

Proto-Oncogene ⟶ Oncogene

2. Abnormal Activity:

Example: Removal of the Regulatory domain in the *Raf gene*
and replaced by *gag gene* ⟶ Raf kinase domain
consciously active ⟶ Cell transformation



RAF PROTO-ONCOGENE

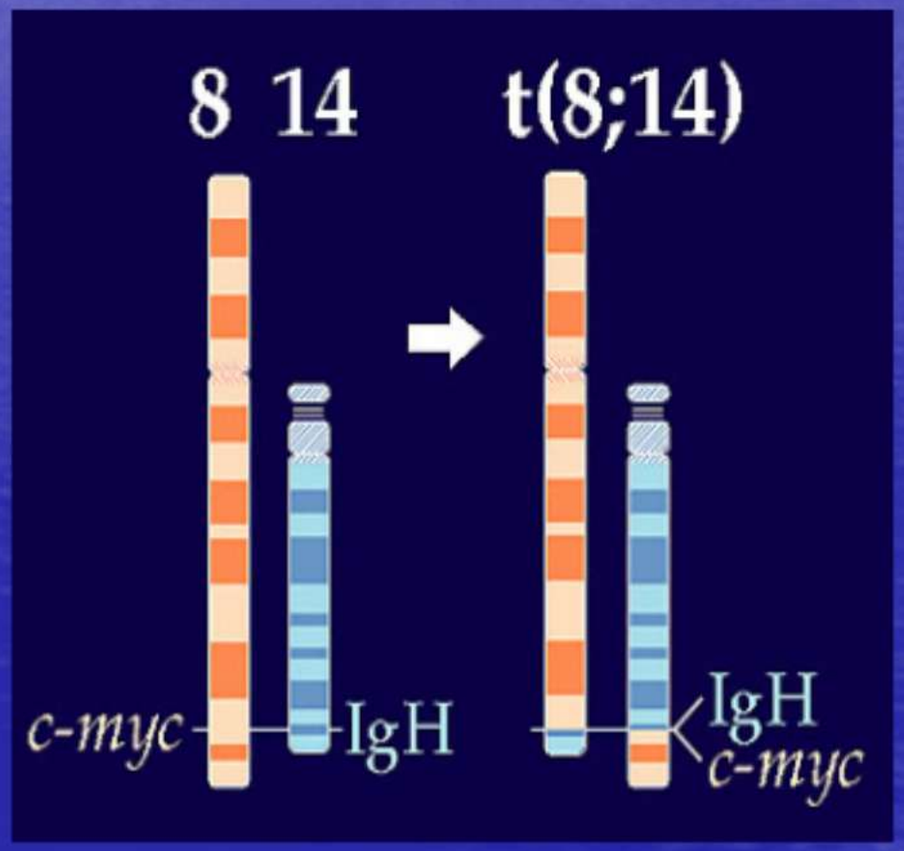


RAF ONCOGENE

Proto-Oncogene ⟶ Oncogene

3. Gene translocation:

Example: *c-myc gene* is translocated from chromosome 8 to the IgH on the chromosome 14 resulting in abnormal c-myc expression ⟶ Cell transformation



Proto-Oncogene ⟶ Oncogene

4. Amplification:

Example: Amplification of *n-myc* ⟶ neuroblastoma.

Amplification of *erbB-2* ⟶ Breast & ovarian carcinomas

How does a Proto-oncogene become an Oncogene?

Proto-Oncogene ⟶ Oncogene

1.Mutation

2. Abnormal Activity

3.Gene Translocation

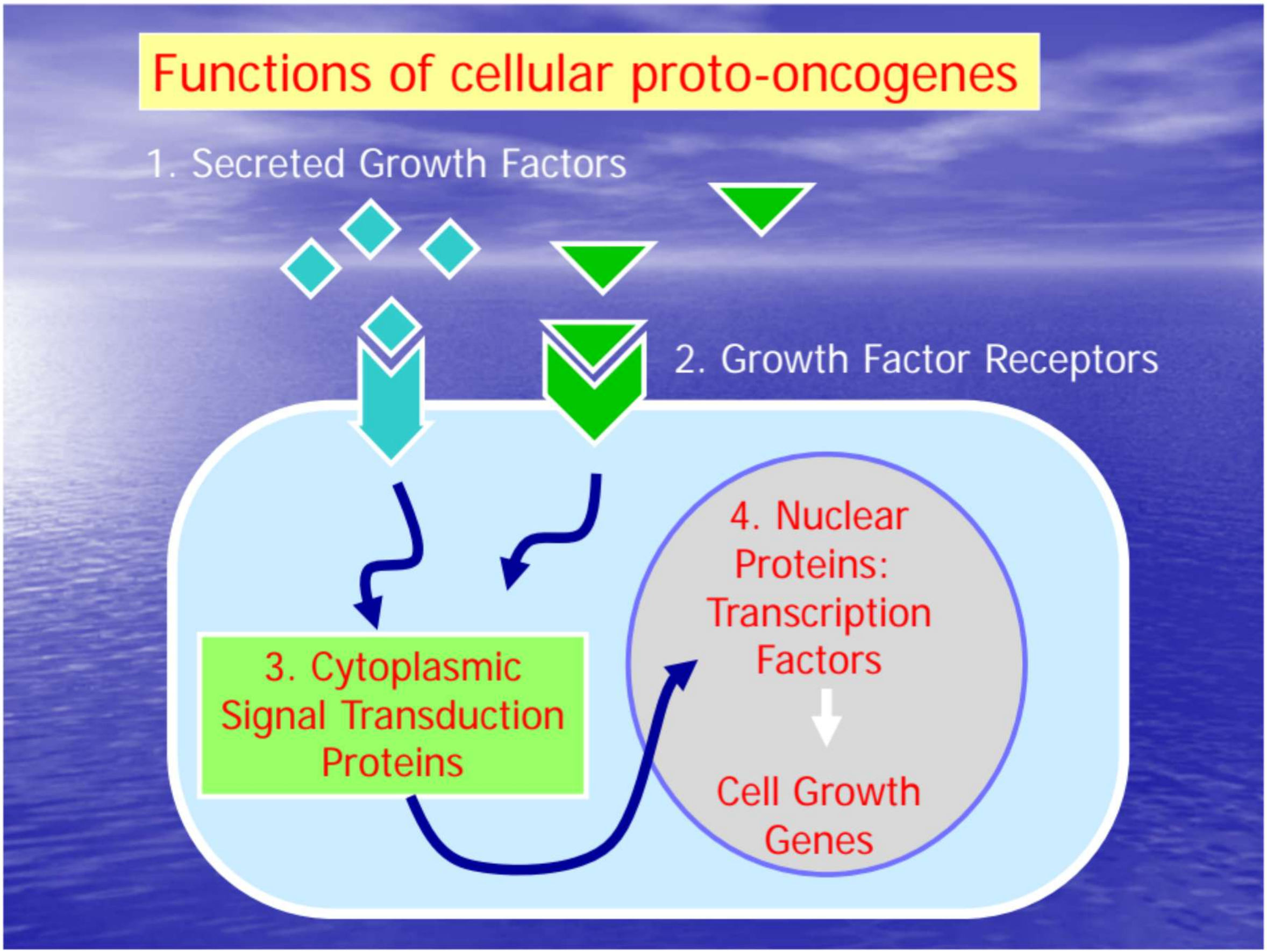
4. Amplification



Functions of proto-oncogene

1. Growth Factor (*example, Epithelium growth factor EGF , and platelet derived growth factor PDGF*)
2. Growth Factor Receptor (*Example; PDGFR*)
3. Signal transduction (*example; Ras, Raf, & MEK*)
4. Transcription Factor (*example; Jun, Fos, Elk-1 & myc*)

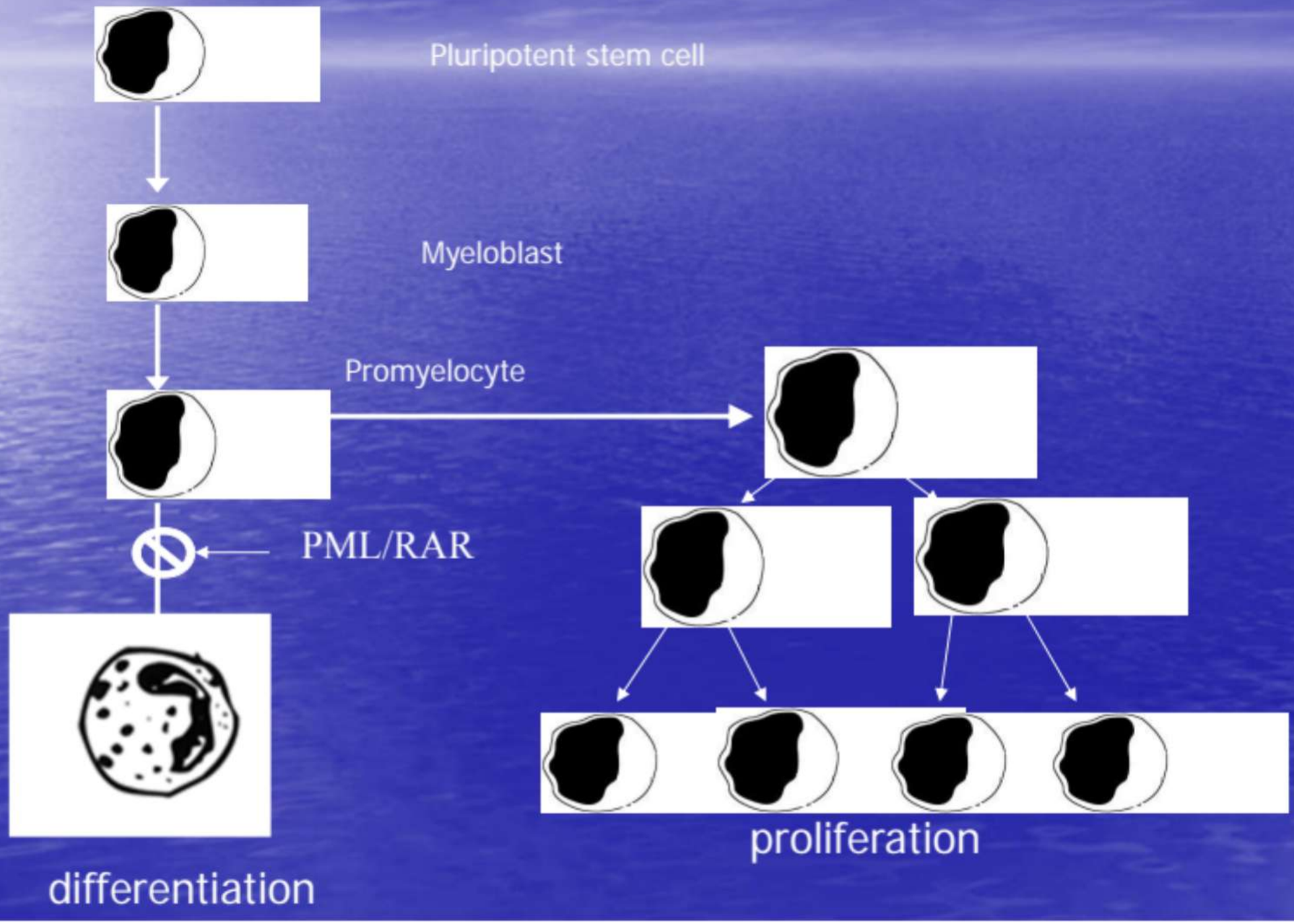
Functions of cellular proto-oncogenes



Oncogenes

- Oncogene causes cancer by affecting:
  1. Cell Proliferation: (example; *Ras, Raf, EGF*)
  2. Cell differentiation (example, *PML/RAR* that inhibits the differentiation of promyelocyte to granulocyte which will maintain the cell in its active proliferate state)
  3. Cell Survival (example; *Pl-3/AKT* which will activate *BCL-2* ⟶ inhibit Apoptosis & maintain cell survival.

PML/RAR Action



Functions of selected proto-oncogenes

Proto-oncogene	Biochemical property
1. Secreted growth factors	
c-sis	Platelet derived growth factor
2. Growth factor receptors	
c-erbB	Epidermal growth factor receptor
3. Signal transduction proteins	
c-abl	Protein kinase
c-src	Protein kinase
H-ras	Small G-protein
K-ras	Small G-protein
4. Nuclear proteins	
c-myc	Transcription factor
c-fos	Transcription factor

Oncogenes in human tumors

Mechanisms of activation of proto-oncogenes

- point mutations
- chromosomal rearrangements or translocations
- gene amplifications

Oncogenes are usually dominant (gain of function)

- cellular proto-oncogenes that have been mutated (and "activated")
- cellular proto-oncogenes that have been captured by retroviruses and have been mutated in the process (and "activated")
- virus-specific genes that behave like cellular proto-oncogenes that have been mutated to oncogenes (i.e., "activated")
- retroviruses has v-onc
- cellular proto-oncogenes are called c-onc

HER2/neu—an oncogene

- A growth factor receptor
- 25-30% of breast cancers over-express HER2/neu
- Which hallmark of cancer does this lead to?
- Herceptin is used as a treatment

TUMOUR SUPRESSOR GENE

Tumour Suppressor Genes

- *Tumour Suppressor genes*: are genes that act to inhibit cell proliferation and tumour development.

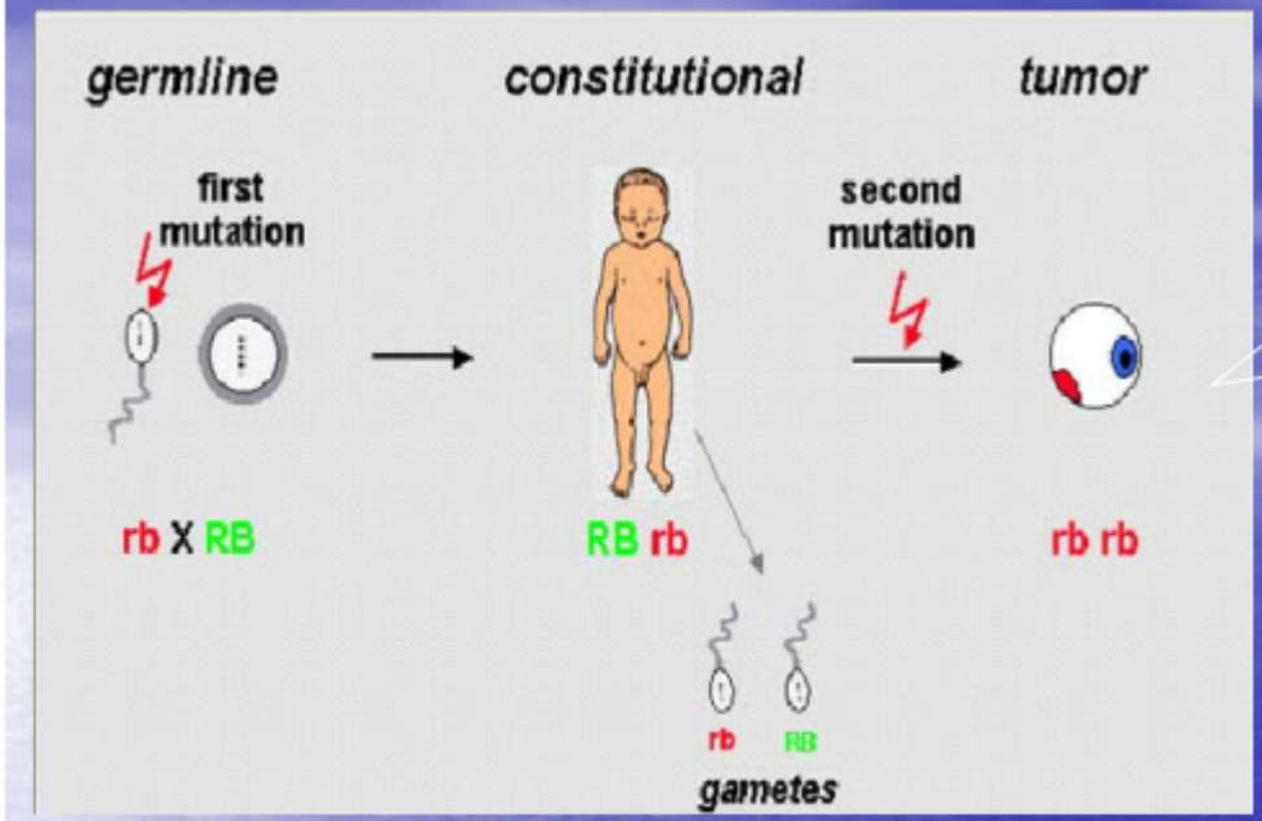
If Tumour Suppressor Gene was

Mutated OR Inactivated

It will lead to cell transformation

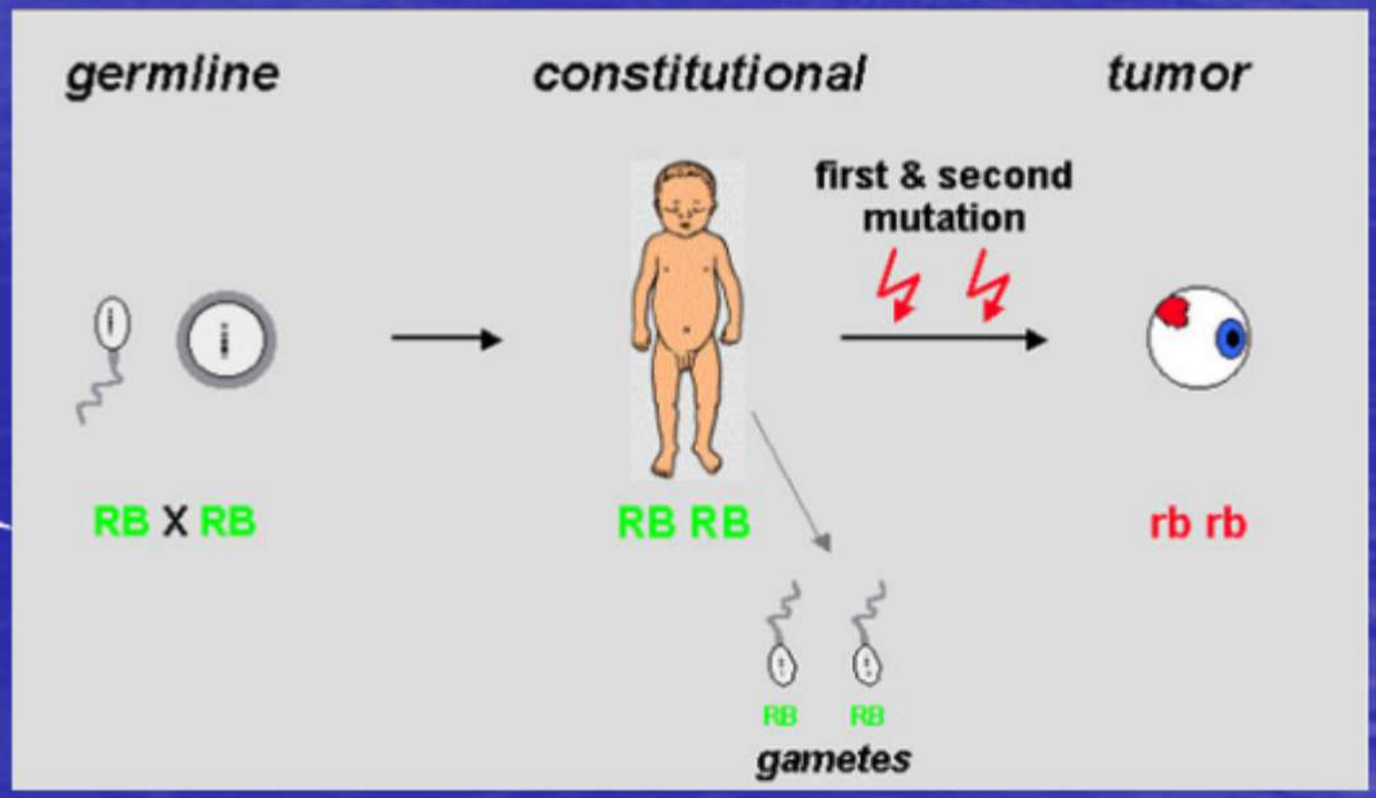
Tumour Suppressor Genes

- Mutation of the tumour suppressor gene will cause cancer.
  - Example; deletion of Rb gene will cause retinoblastoma. The development of retinoblastoma can be either:
    - Hereditary: a defective copy of Rb gene is inherited from the affected parents.
    - Nonhereditary: in which 2 normal Rb genes are inherited and develop mutation during life.
  - Retinoblastoma is developed if 2 somatic mutations inactivate both copies of Rb in the same cell.



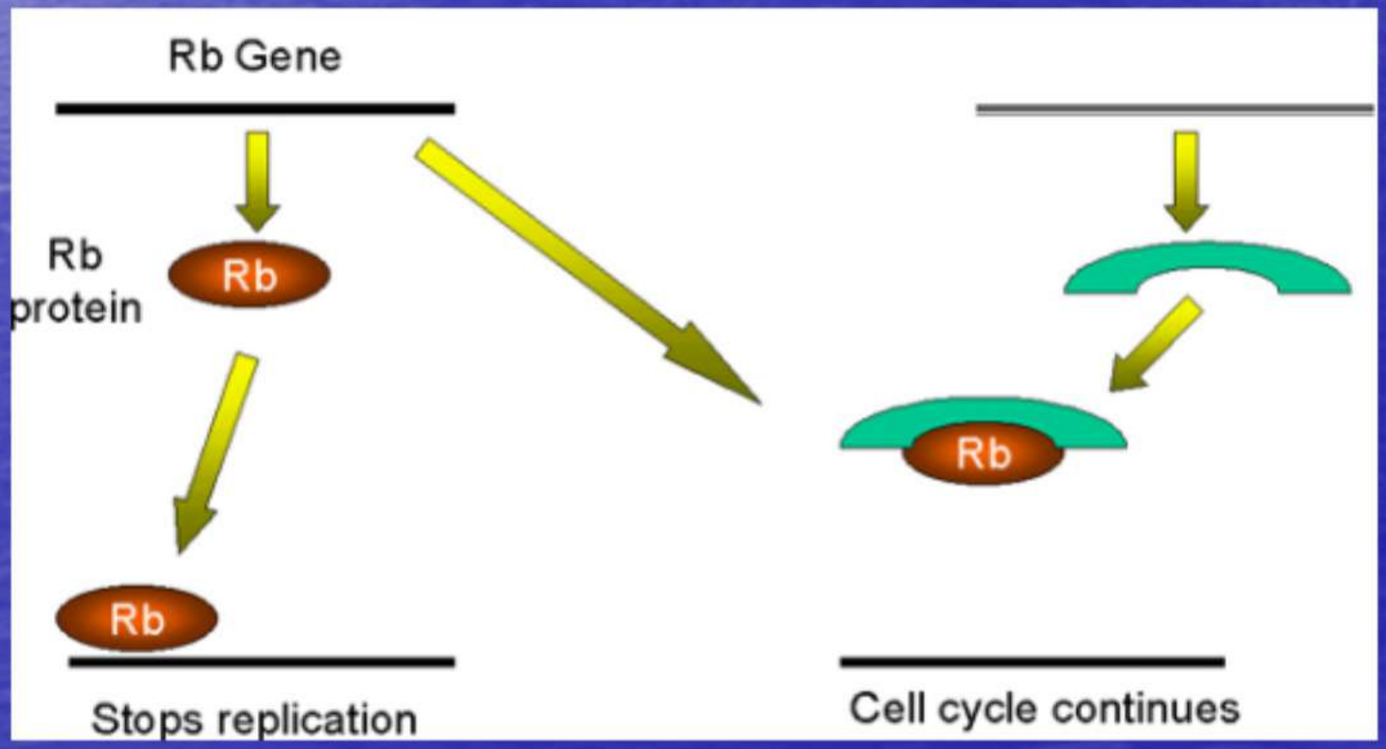
Hereditary Mutation

Non-hereditary Mutation



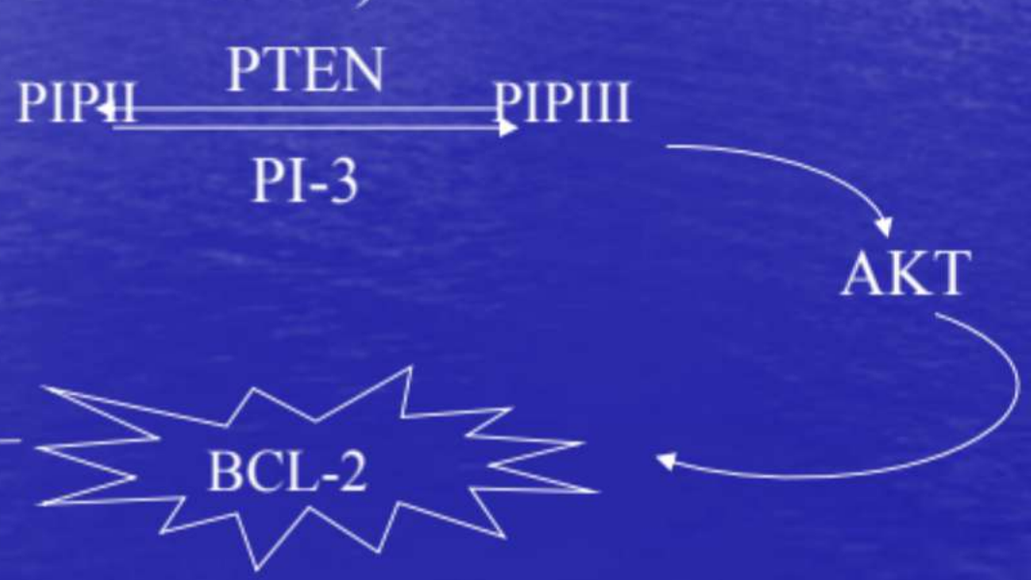
Tumour Suppressor Genes

- Inactivation of Tumour suppressor gene will cause cancer.
  - If the Rb gene interact with DNA tumour virus (SV40) it will induce cell transformation.



Function of Tumour Suppressor gene

1. *Antagonize the action of oncogene*. (ex.PTEN which converts PIPIII to PIPII because PIPIII will activate Pl-3/AKT which will activate BCL-2 that will inhibit apoptosis and induce cell transformation)



Function of Tumour Suppressor gene

2. *Transcription factors*

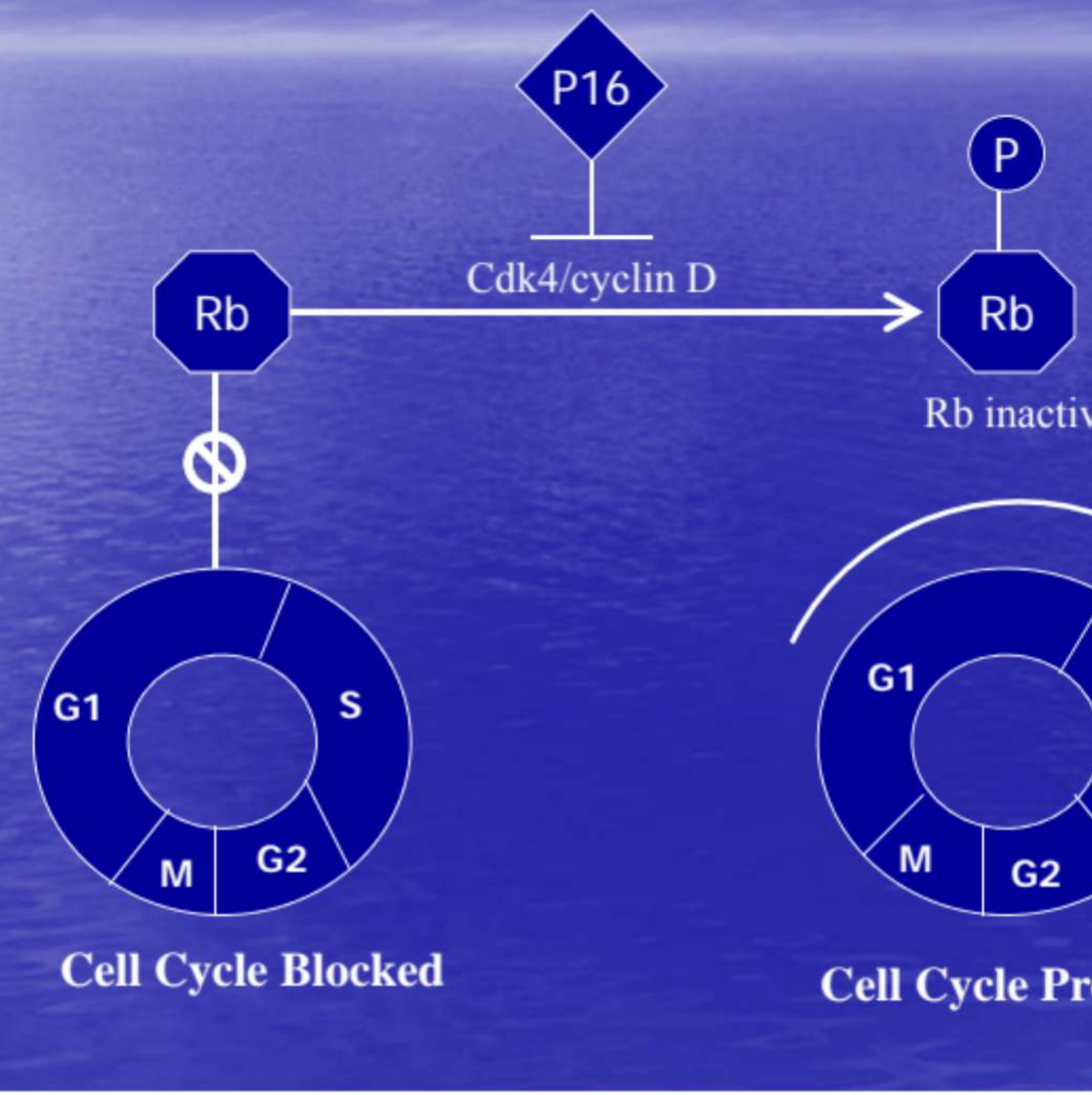
- Repressor transcription factors: example; WT1 is a repressor that appears to suppress transcription factor (Insulin like growth factor) which will contribute in the development of tumour.
- Activator transcription factors: example; SMAD family that are activated by TGF-β, leading to inhibition of cell proliferation.

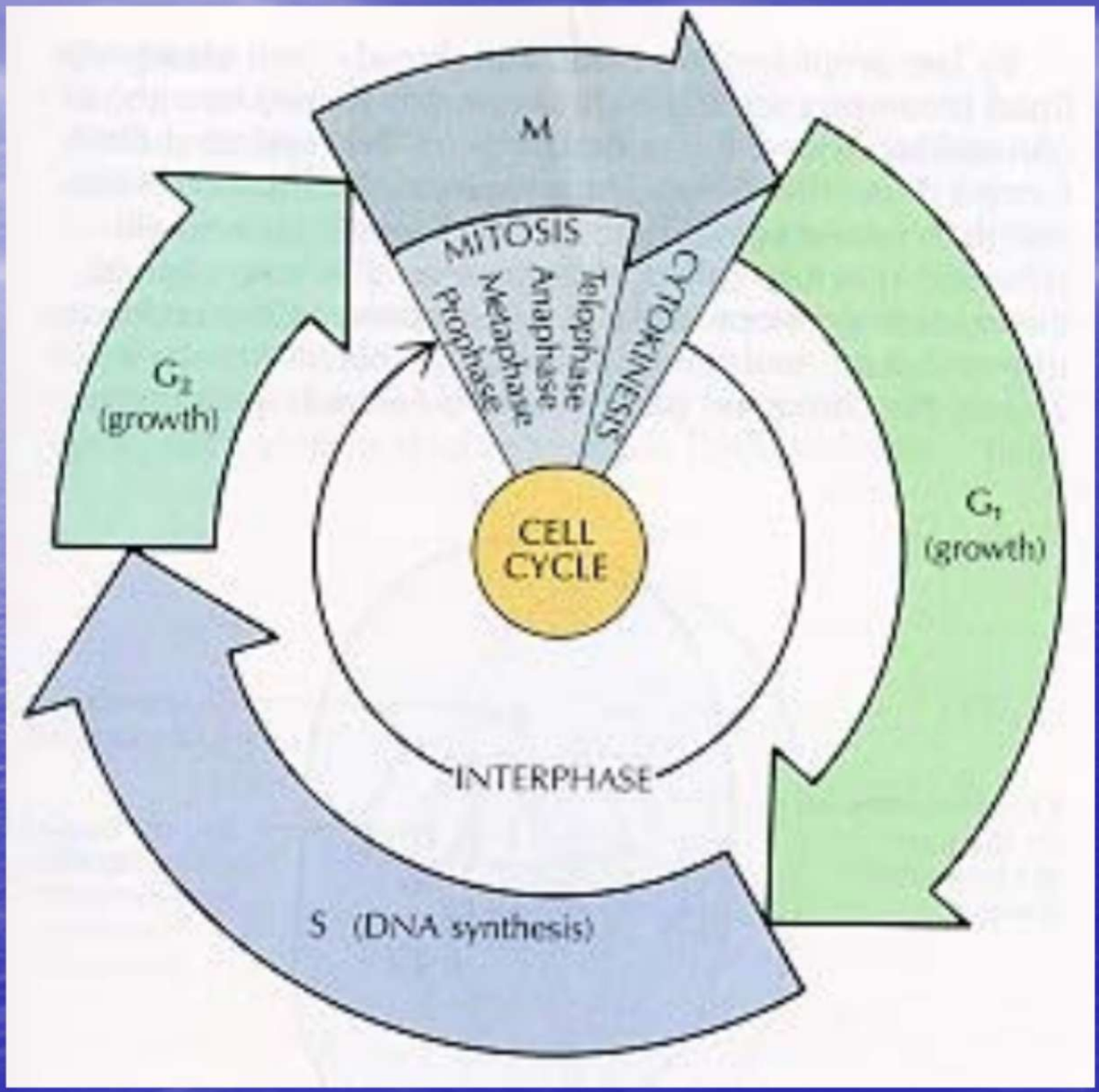
Function of Tumour Suppressor gene

3. Regulate cell cycle :

- *Rb gene:* that inhibits the cell cycle in the G1 phase → decrease cell proliferation.
- *INK-4 gene:* that produces P16 → that inhibits cdk4/cyclin D action (to phosphorylate Rb gene to inactivate it's action)
- *P53:* that produces P21 that has the same action of P16 in inhibiting the action of cdk4/cyclin D

Regulate cell cycle

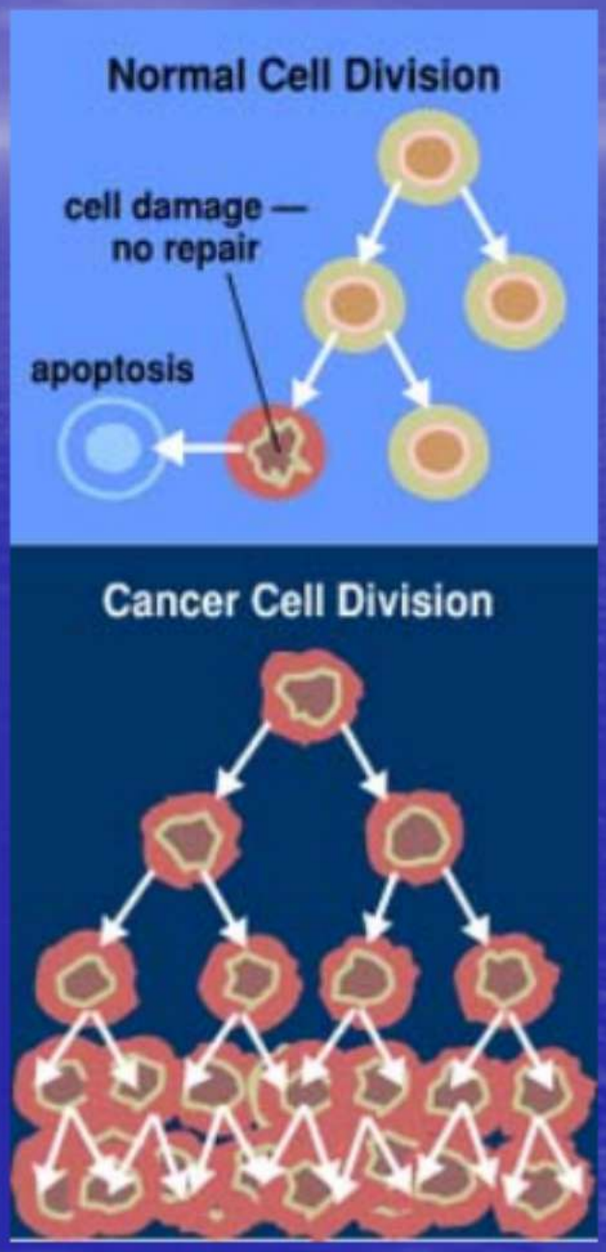



M
MITOSIS
Telophase
Anaphase
Metaphase
Prophase
CYTOKINESIS
G2
(growth)
CELL
CYCLE
G1
(growth)
INTERPHASE
S (DNA synthesis)

Role of Cell Division in Cancer

Top = normal cell division

Bottom = unregulated cell division and tumor formation



Malignant

If tumor invades surrounding tissue (cancerous)

Benign

If tumor has no effect on surrounding tissue (non-cancerous)

Metastatic

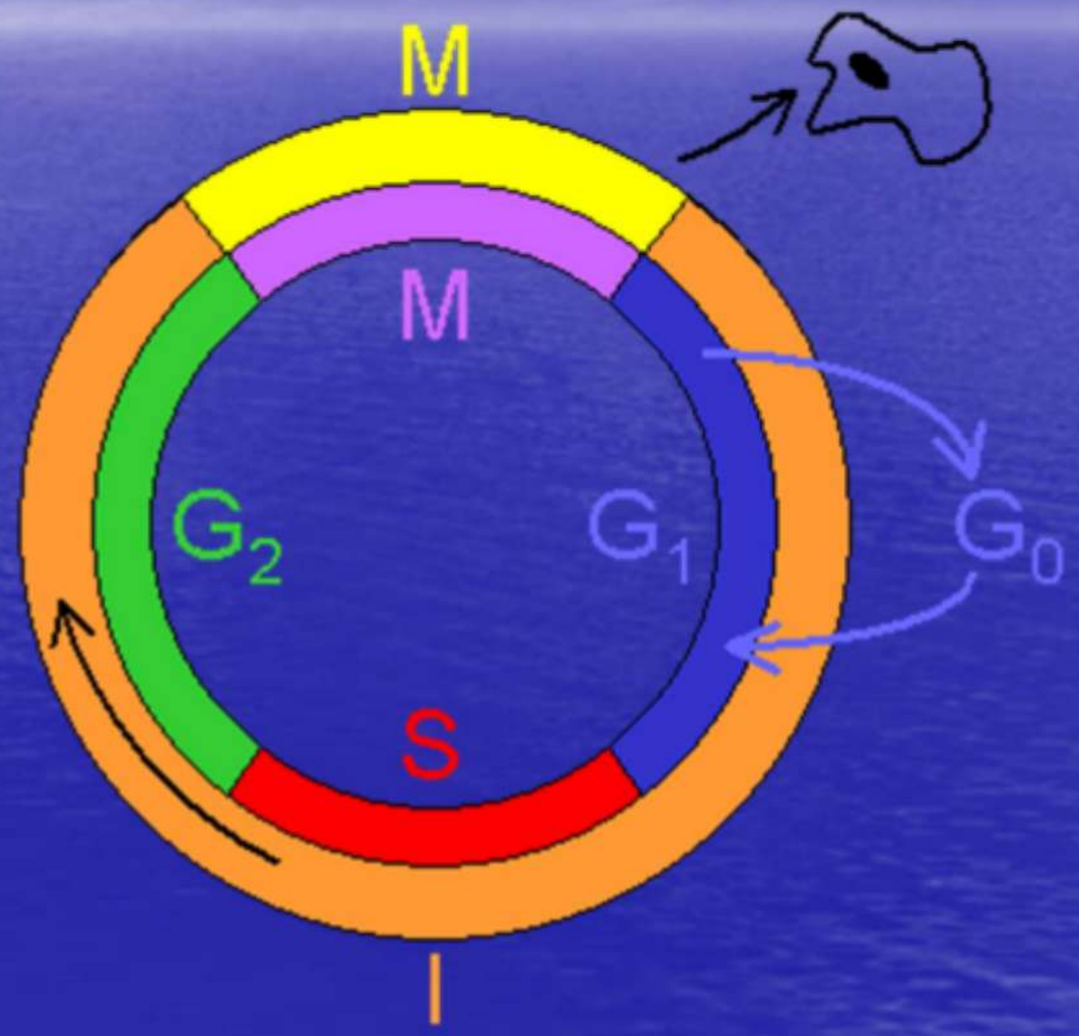
If individual cells break away and start a new tumor elsewhere (cancerous)

Image from the National Cancer Institute

Review: The cell cycle has four phases and controls cell division

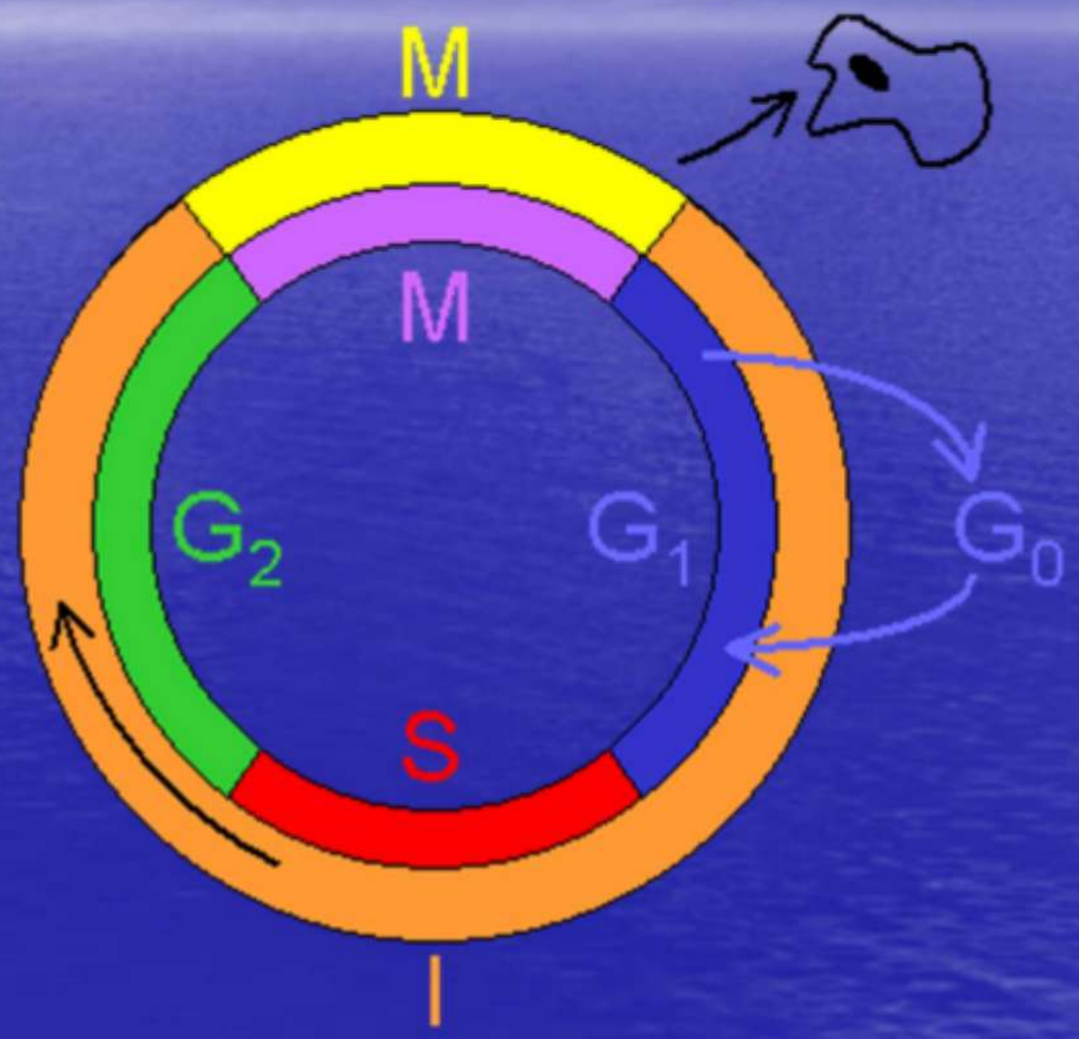
- Two gap or growth phases (G1 and G2)
- S phase - DNA synthesis
- M phase - Mitosis

Interphase



Cell Cycle Checkpoints

- Three checkpoints in cell cycle
  - **G1-S transition**
  - **G2-M transition**
  - **Exit M phase transition**
- Checkpoints are where the cell assesses whether conditions are favorable for cell division.
  - When the environment is not favorable (for example, when the cell's DNA is damaged), a protein called **p53** can stop the cell cycle and cause the cell to die.
  - When the proteins that regulate the cell cycle are mutated or absent, cells can divide uncontrollably, leading to cancer.



Cell-cycle dependent phosphorylation of Rb

Phosphorylation of Rb allows cells to transit the restriction point and enter S phase

Hyperphosphorylated Rb

Restriction point

S phase

G1 phase

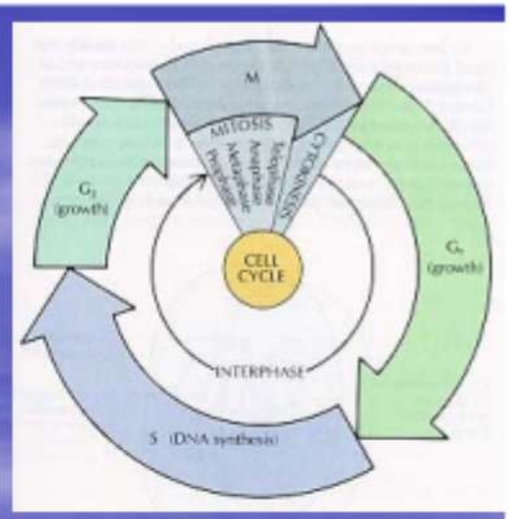
G2 phase

M phase

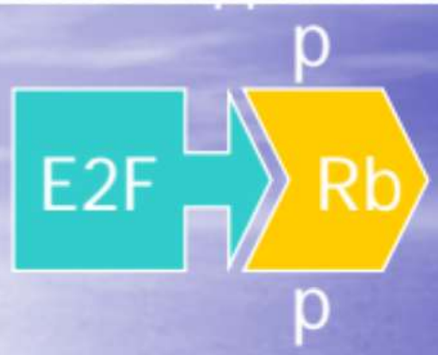
G0
Quiescent cells

Hypophosphorylated Rb

Function of the Rb protein



Growth suppression



Relief of growth suppression



G1 phase phosphorylation releases E2F



Adenovirus E1A oncoprotein binding releases E2F



Gene mutation affecting binding pocket releases E2F

- E2F is a transcription factor that mediates growth-dependent activation of genes required to make the transition into and through S phase

- Rb binds and inactivates E2F under conditions of growth suppression

- There are several ways to alleviate growth suppression resulting in controlled or uncontrolled cell growth

Function of Tumour Suppressor gene

4. Induce apoptosis:

- *P53* release will → increase Bax → form holes in the mitochondria → release cytochrom c activate apoptosis →

p53—a classic tumor suppressor

- "The guardian of the genome"
- Senses genomic damage
- Halts the cell cycle and initiates DNA repair
- If the DNA is irreparable, p53 will initiate the cell death process

Rb—a classic tumor suppressor

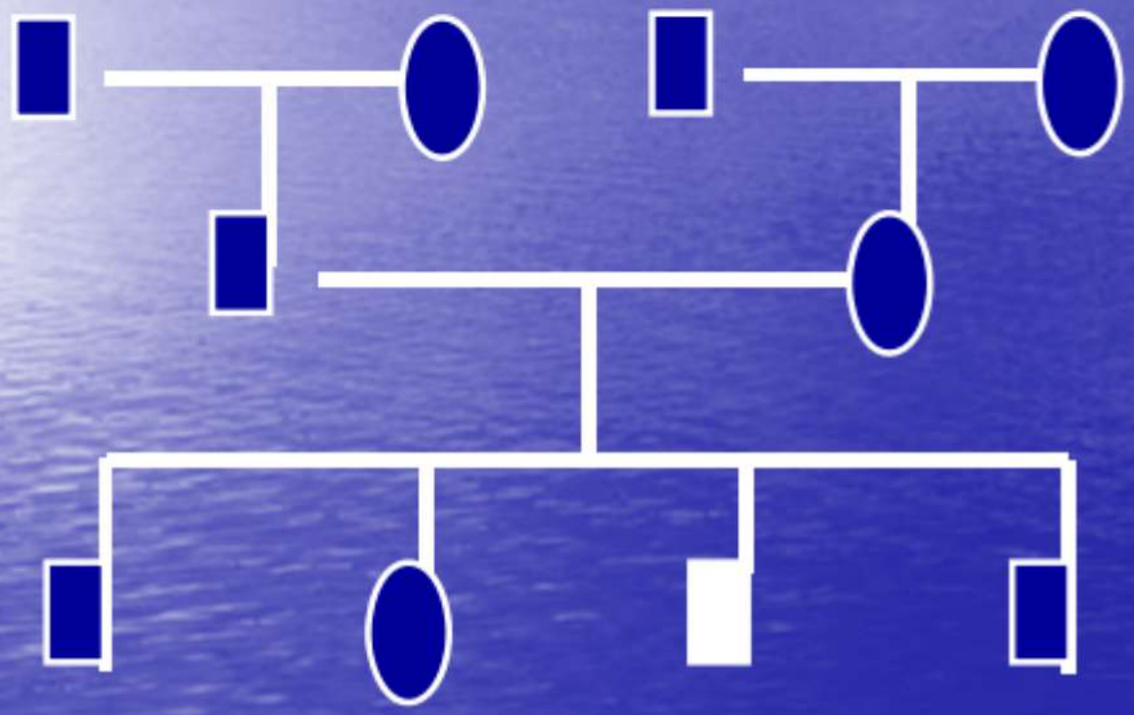
- Rb binds to a protein called E2F1
- E2F1 initiates the G1/S cell cycle transition
- When bound to Rb, E2F1 can't function
- Thus, Rb is a crucial cell cycle checkpoint

Tumour suppressors

- "Guardian(s) of the genome"
- Often involved in maintaining genomic integrity (DNA repair, chromosome segregation)
  - Mutations in tumor suppressor genes lead to the "mutator phenotype"—mutation rates increase
  - Often the 1st mutation in a developing cancer

Sporadic (Mendelian) forms of cancer
Knudson's two-hit hypothesis

Sporadic



Normal tumor suppressor gene

Somatic mutation in one allele

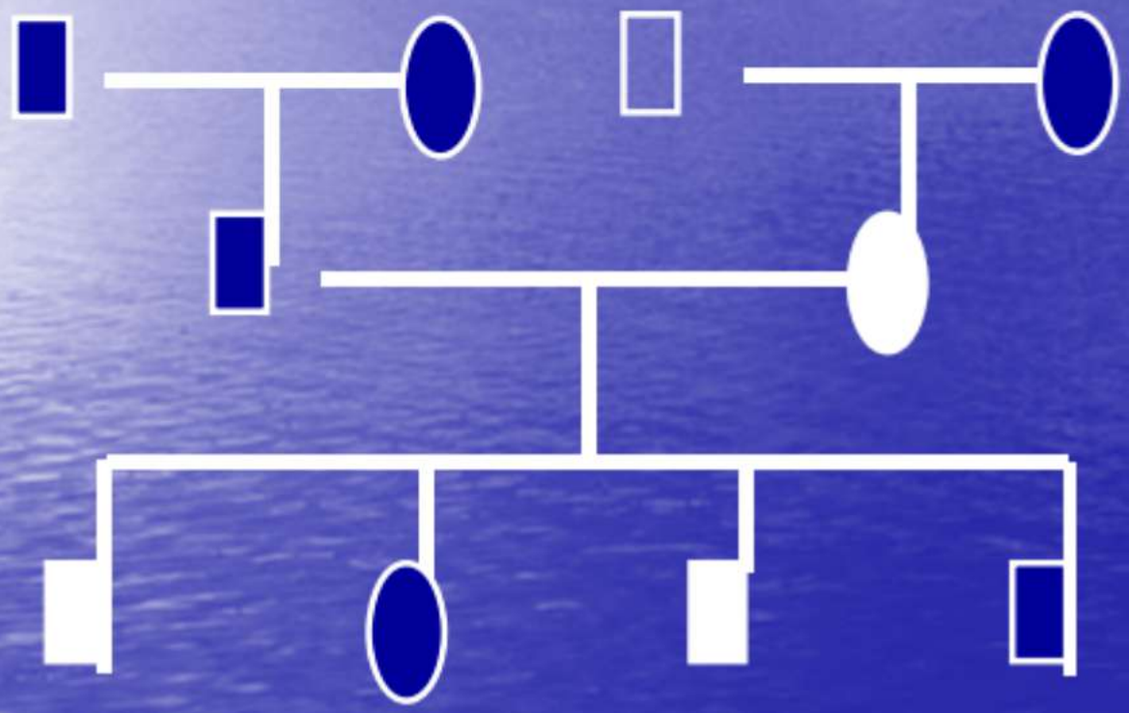
Somatic mutation in other allele

Single tumors, unilateral, later-onset

- two mutations (two hits) are required for loss of tumor suppressor function

Familial (Mendelian) forms of cancer
Knudson's two-hit hypothesis

Familial



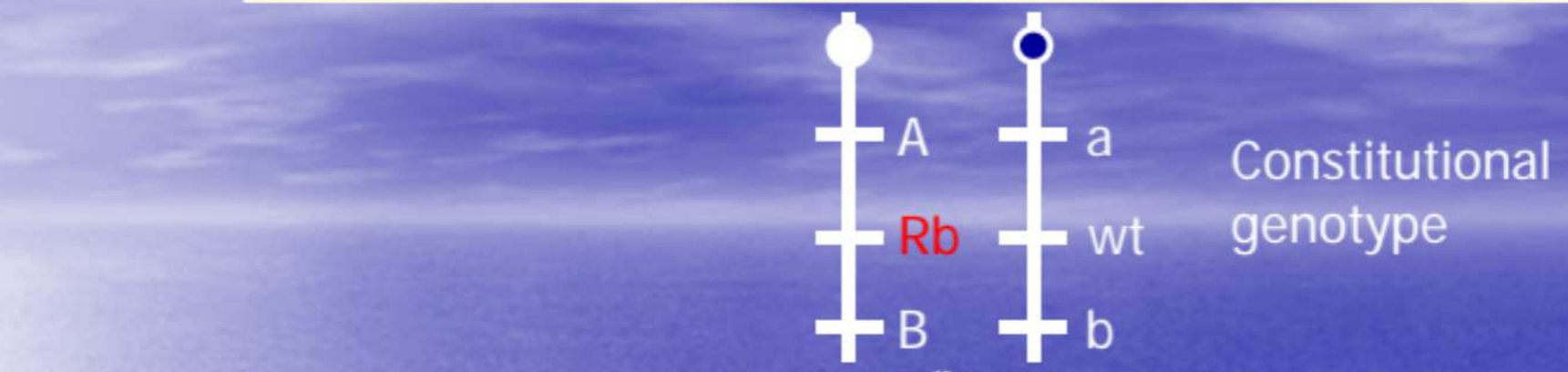
Tumor suppressor gene containing a germline mutation in one allele - heterozygous for the mutation

Somatic mutation in other allele

Multiple tumors, bilateral, early-onset

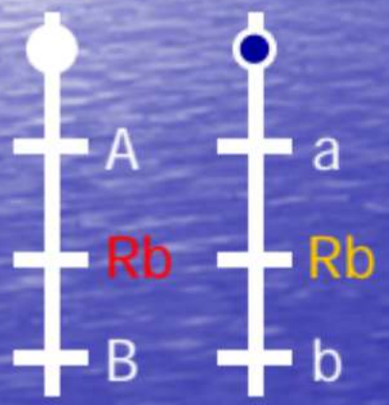
- two mutations (two hits) are required for loss of tumor suppressor function
- the first "hit" is inherited and the second "hit" is somatic

Chromosomal mechanisms that could lead to loss of function due to loss of heterozygosity

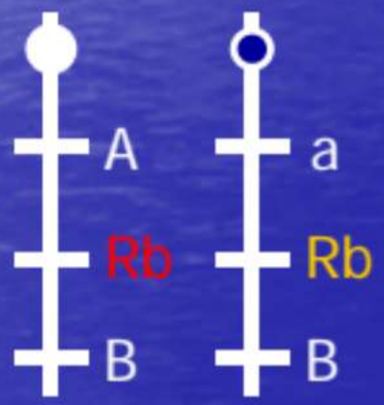


Constitutional genotype

Somatic mutational events - the second "hit"



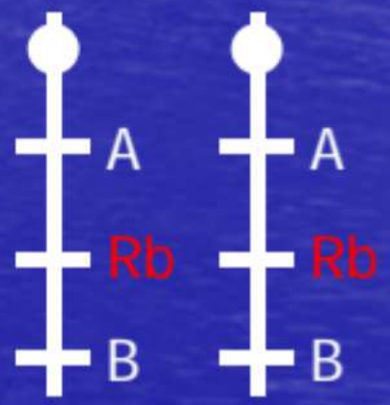
Local mutational events



Somatic recombination



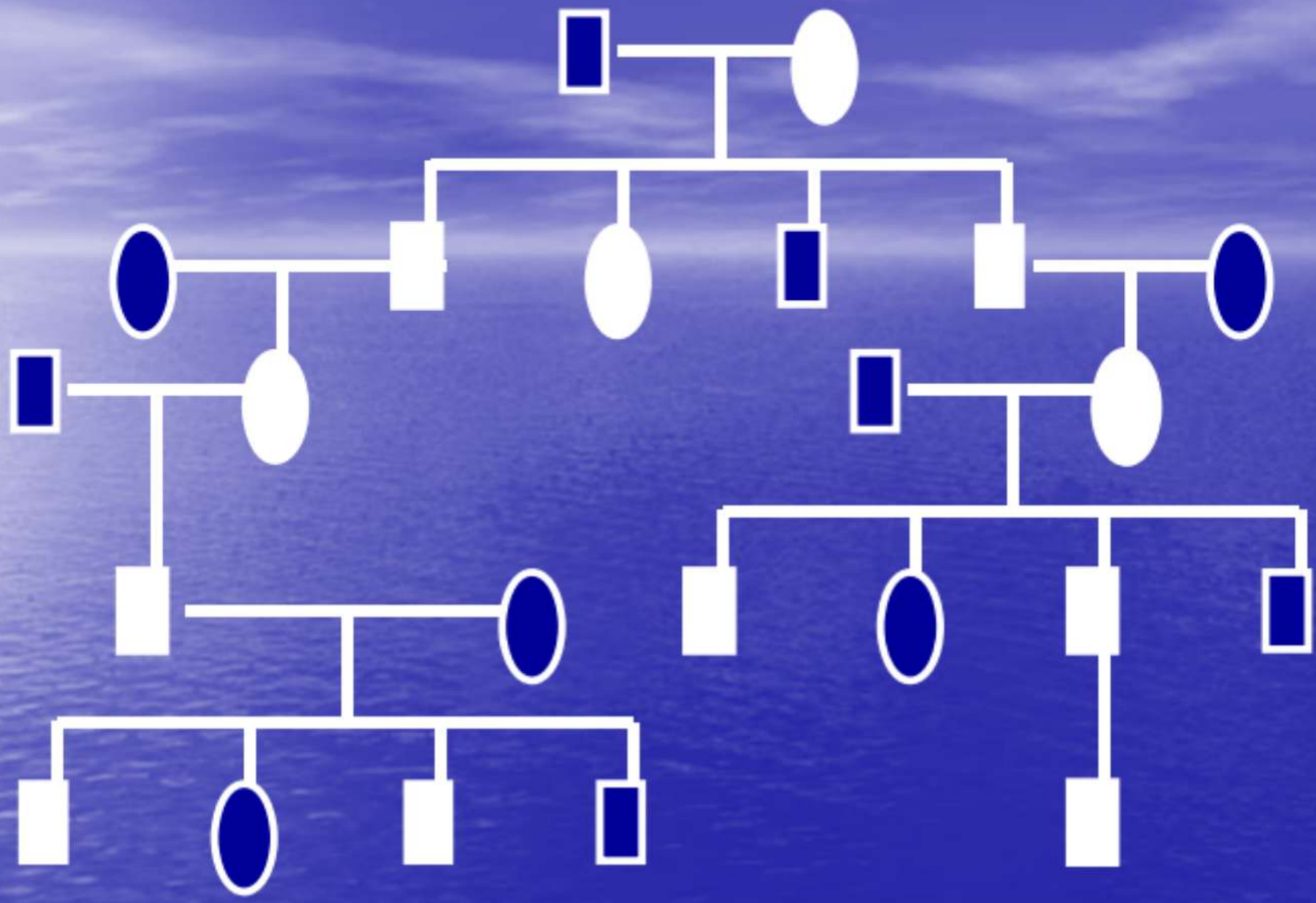
Chromosome loss



Chromosome loss and duplication

Tumor genotypes in retinoblastoma

Li-Fraumeni syndrome - caused by mutations in the p53 gene



Breast cancer

Sarcoma

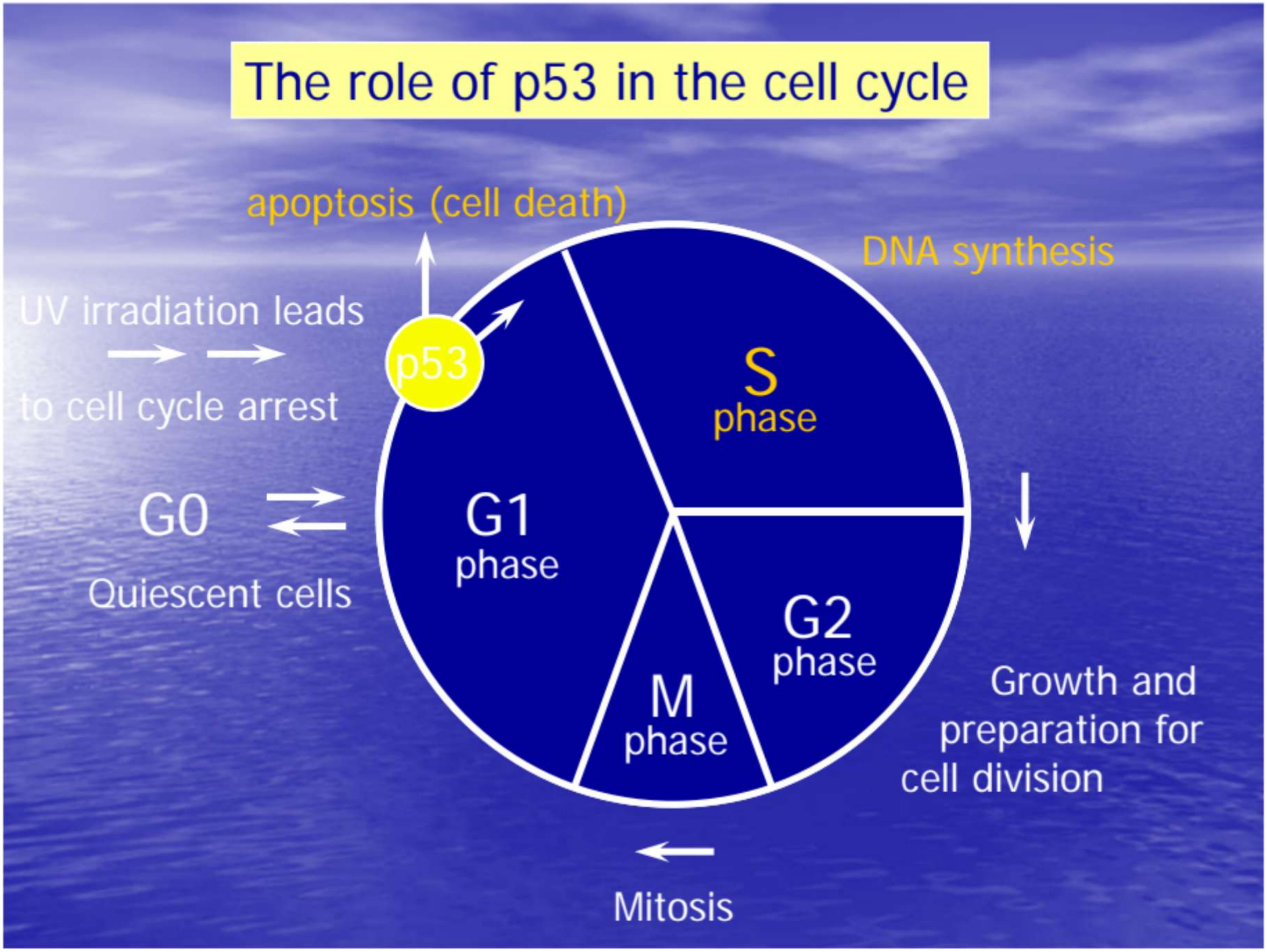
Other malignant neoplasms

There are multiple neoplasms in Li-Fraumeni families that are inherited in an autosomal dominant fashion

p53 is the "guardian of the genome"

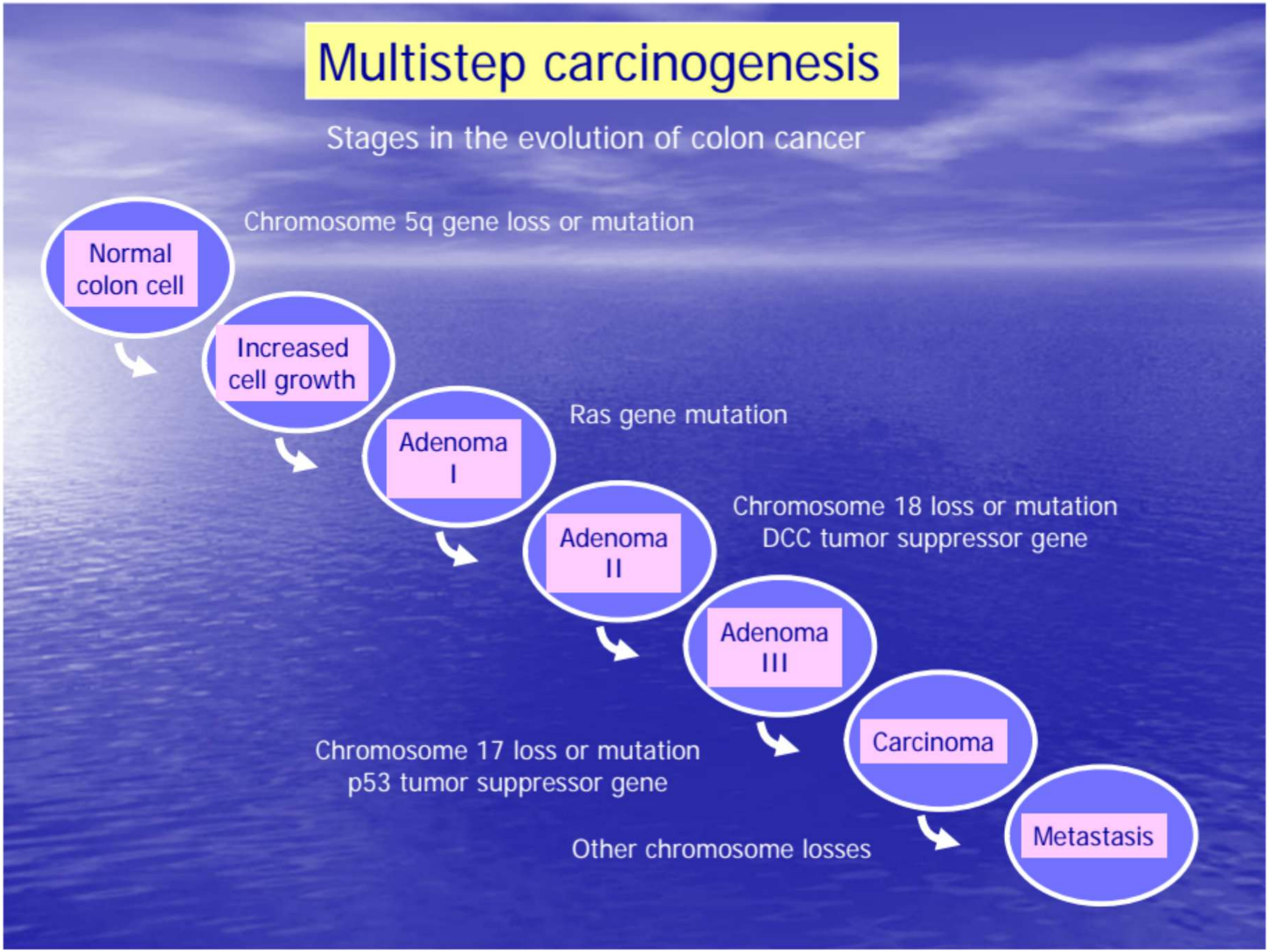
- germline p53 mutations are found in Li-Fraumeni syndrome
- p53 is frequently found mutated in human tumors
- the p53 protein functions as a transcription factor that regulates cell-cycle and DNA repair genes
- UV irradiation causes cell-cycle arrest in G1 that is dependent on p53; cells that contain a mutated p53 cannot arrest and go into S phase and replicate damaged DNA
- p53 loss-of-function mutations result in the replication of cells with damaged DNA and to the further accumulation of other mutations affecting oncogenes and tumor suppressor genes, and to an increased likelihood of cancer

The role of p53 in the cell cycle



Multistep carcinogenesis

Stages in the evolution of colon cancer



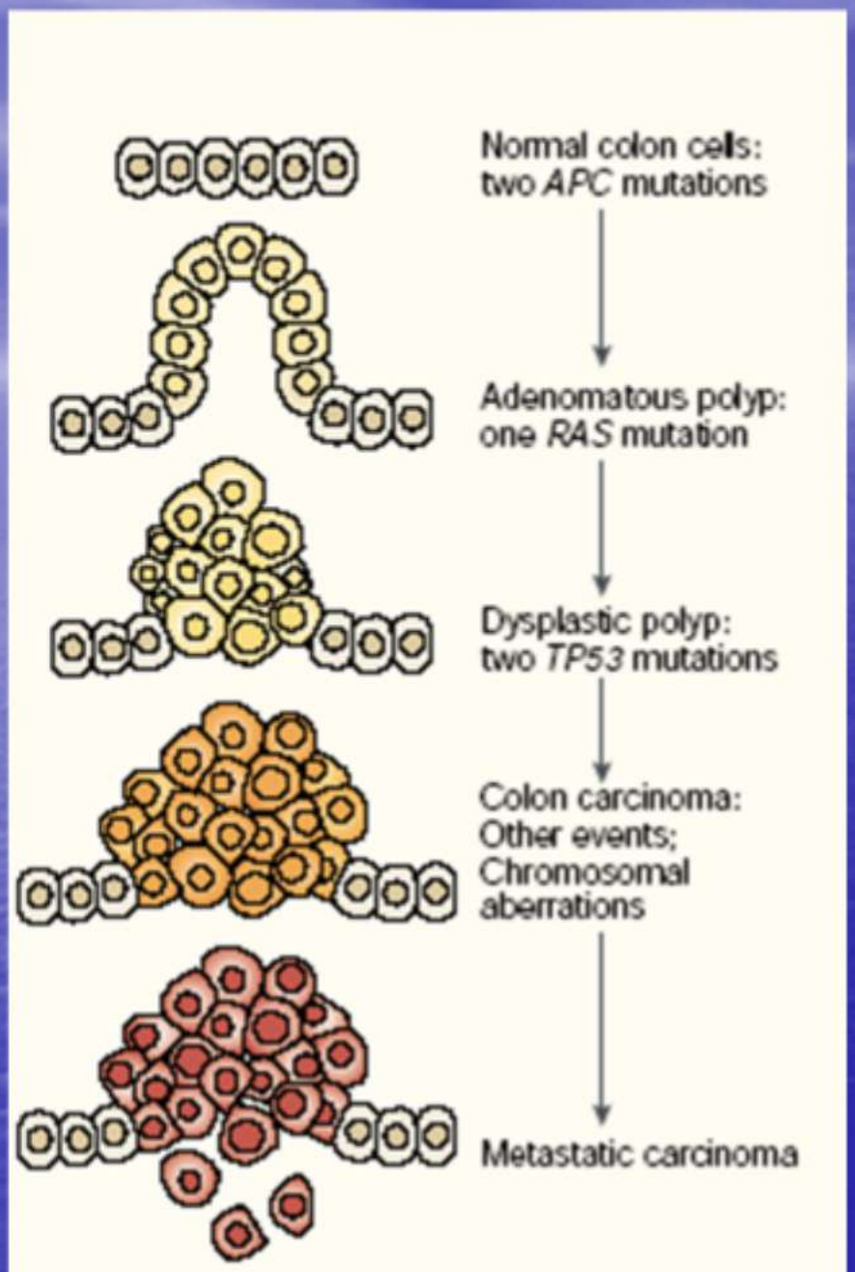
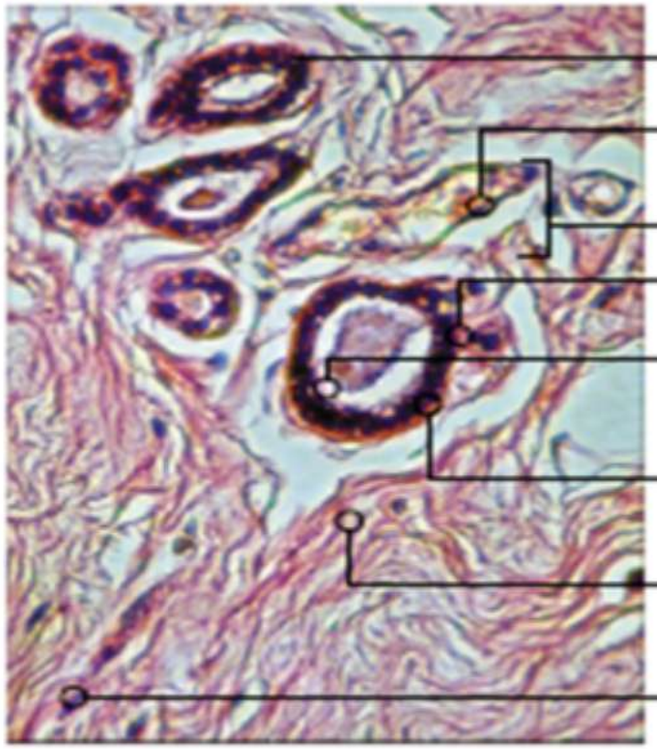
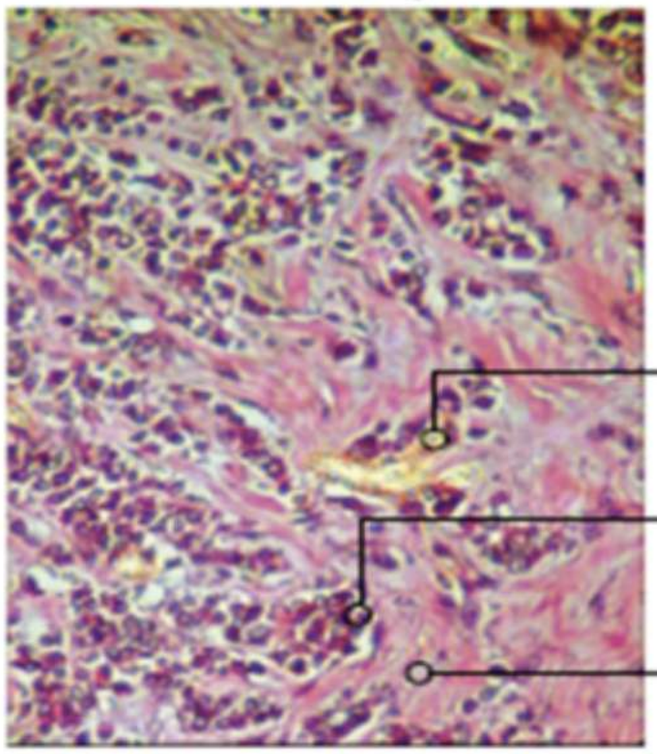


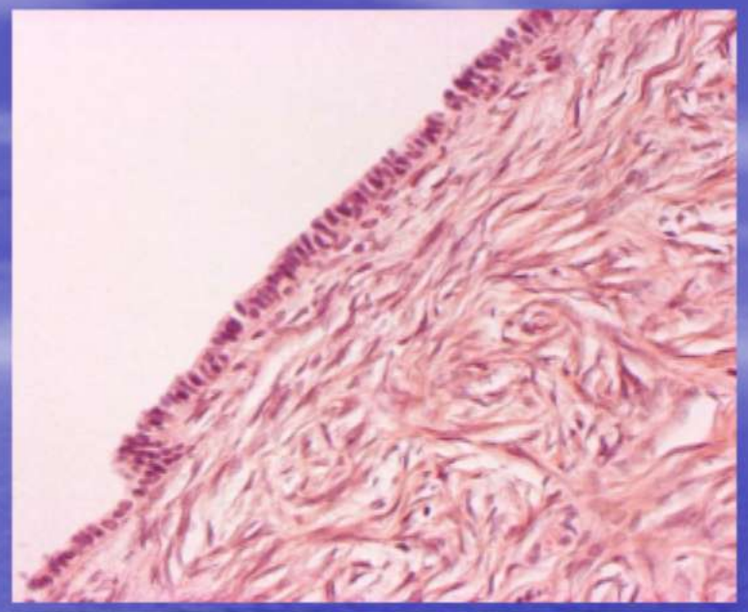
Figure 5 | **A possible five-hit scenario for colorectal cancer, showing the mutational events that correlate with each step in the adenoma–carcinoma sequence.** Based on a model from Fearon and Vogelstein (REF. 47).

a Normal mammary gland

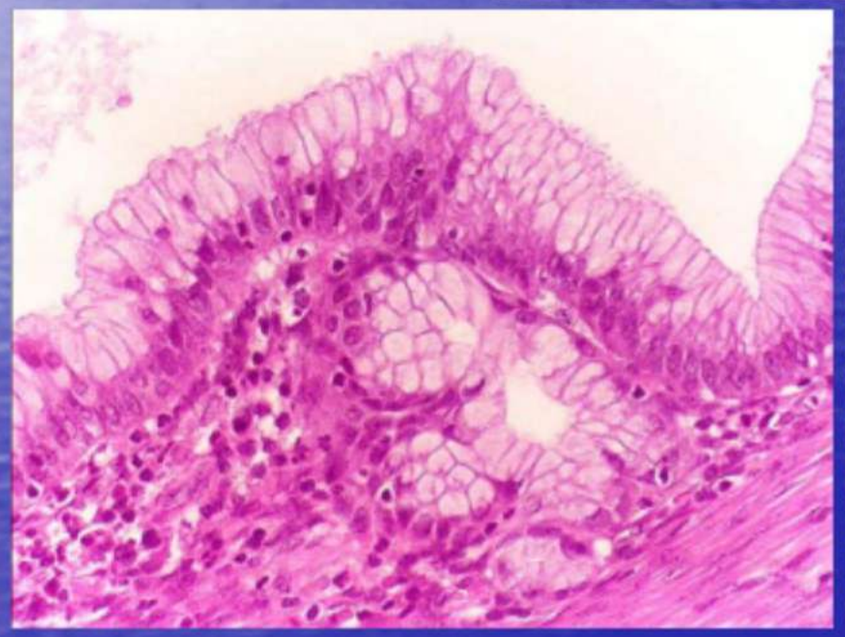


b Lobular mammary carcinoma

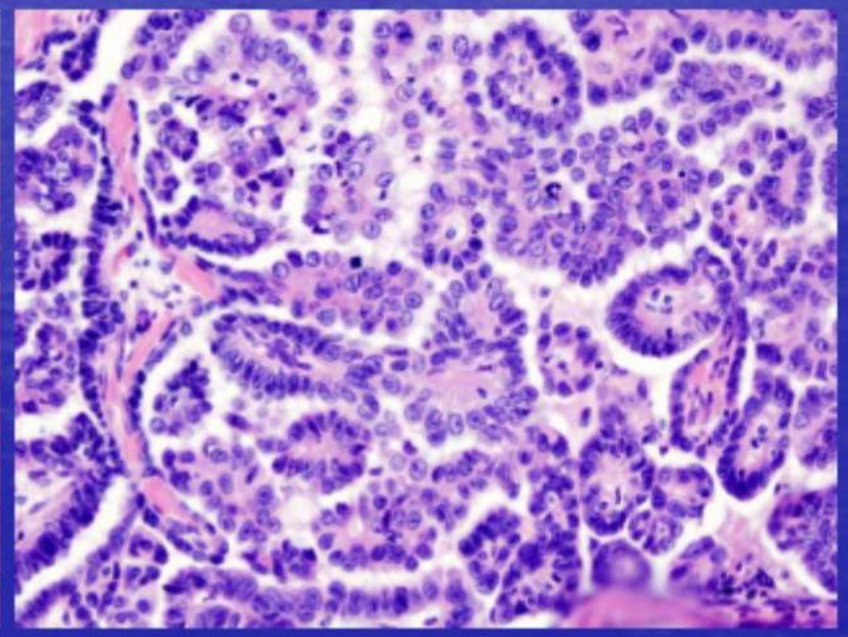




Normal ovarian epithelium



Ovary cystoadenoma (benign)



Ovarian adenocarcinoma (malignant)

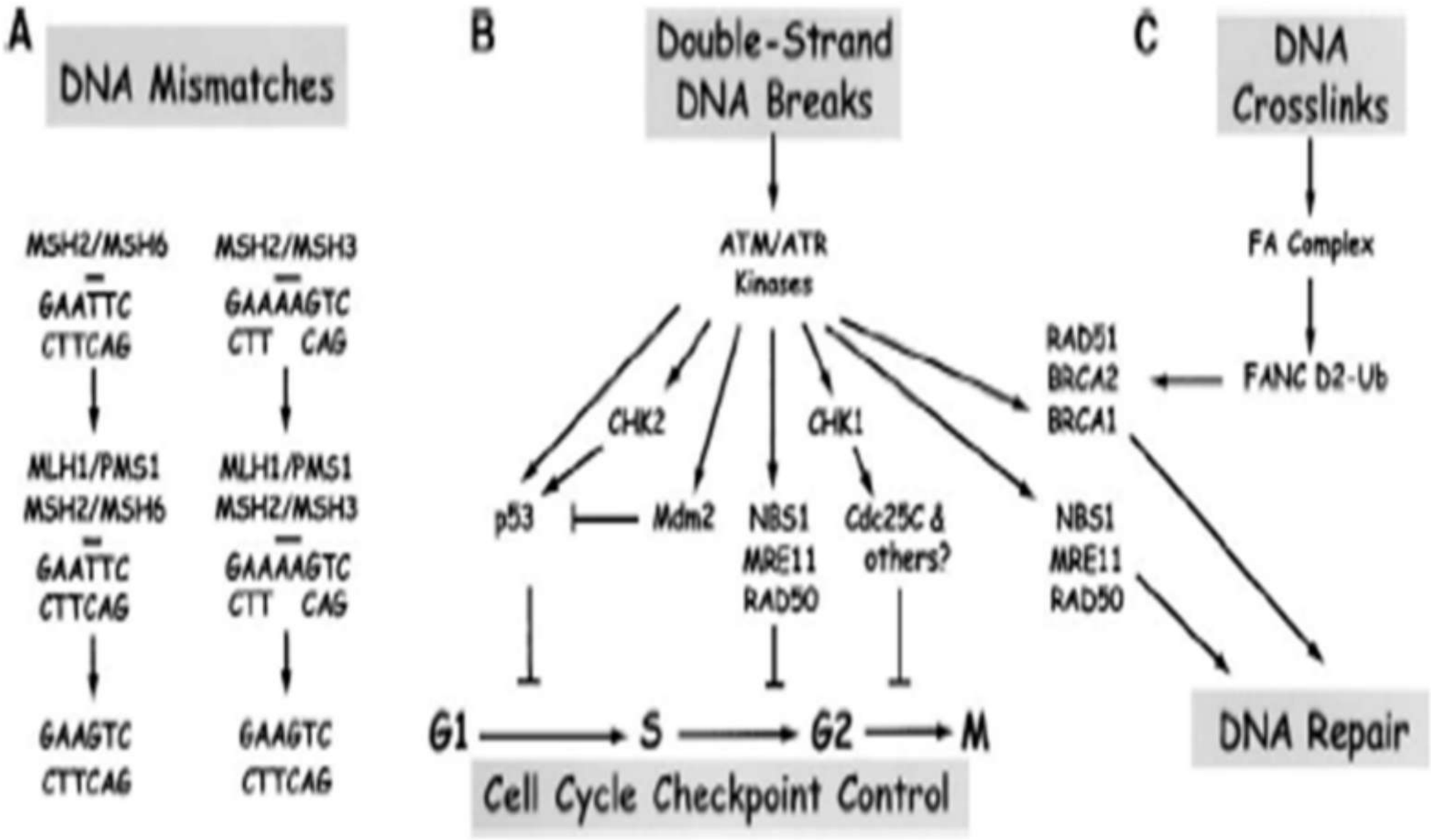
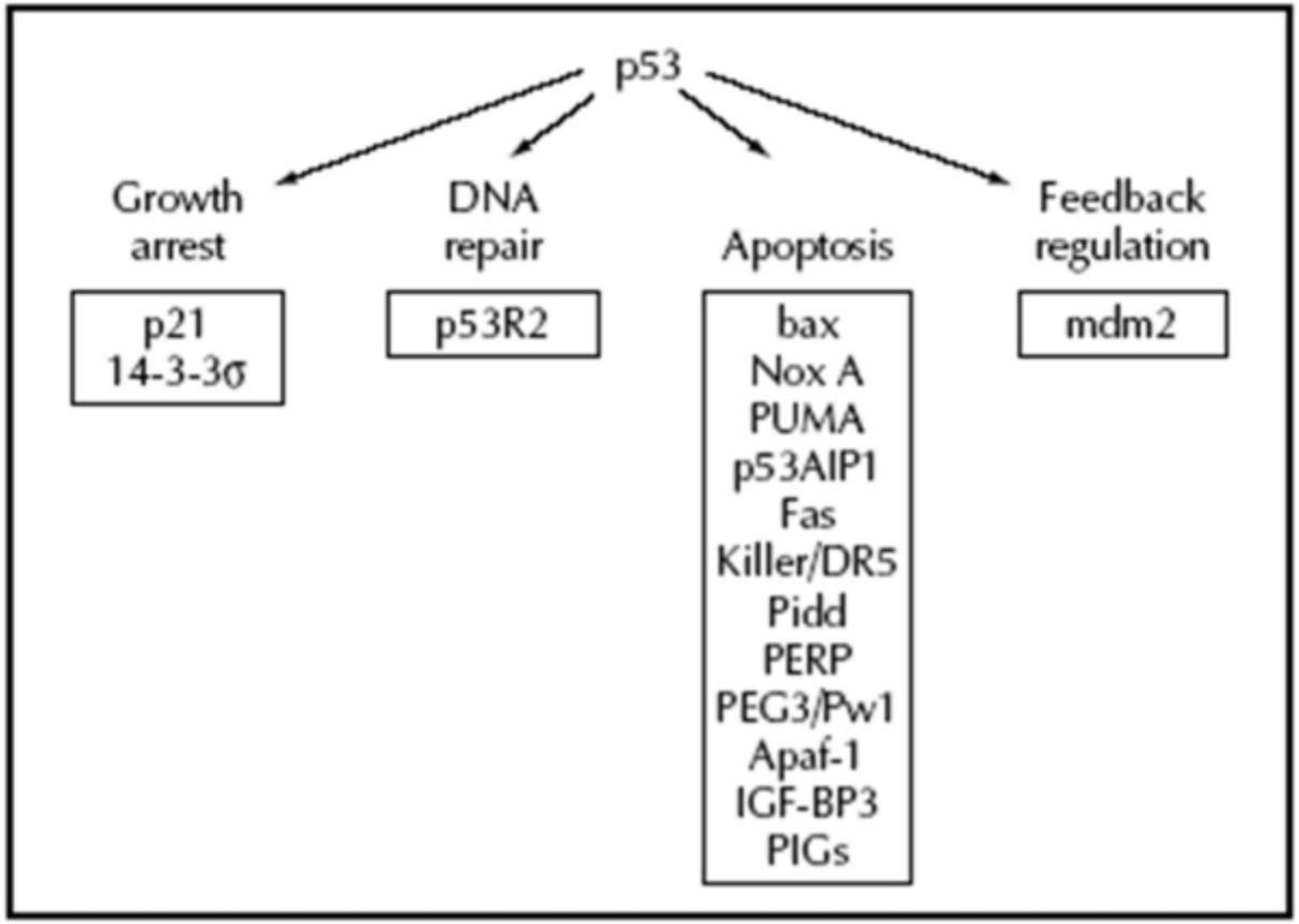

A
DNA Mismatches
MSH2/MSH6
GAATTC
CTTCAG
MSH2/MSH3
GAAAAGTC
CTT CAG
MLH1/PMS1
MSH2/MSH6
GAATTC
CTTCAG
MLH1/PMS1
MSH2/MSH3
GAAAAGTC
CTT CAG
GAAGTC
CTTCAG
GAAGTC
CTTCAG
B
Double-Strand
DNA Breaks
ATM/ATR
Kinases
CHK2
CHK1
p53
Mdm2
NBS1
MRE11
RAD50
Cdc25C &
others?
G1
S
G2
M
Cell Cycle Checkpoint Control
C
DNA
Crosslinks
FA Complex
FANC D2-Ub
RAD51
BRCA2
BRCA1
NBS1
MRE11
RAD50
DNA Repair

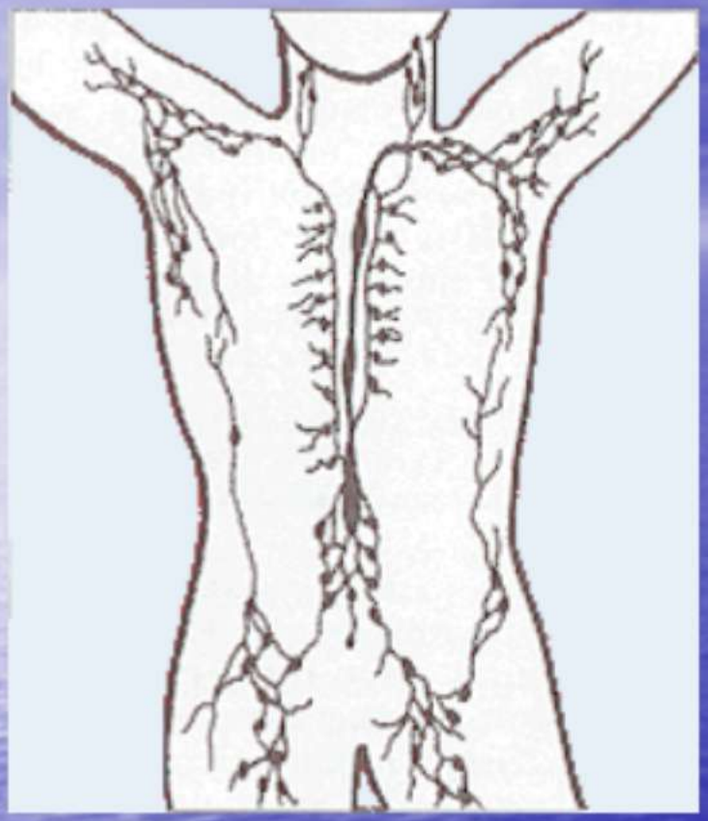
Figure 3. Target genes in cellular responses to *p53*

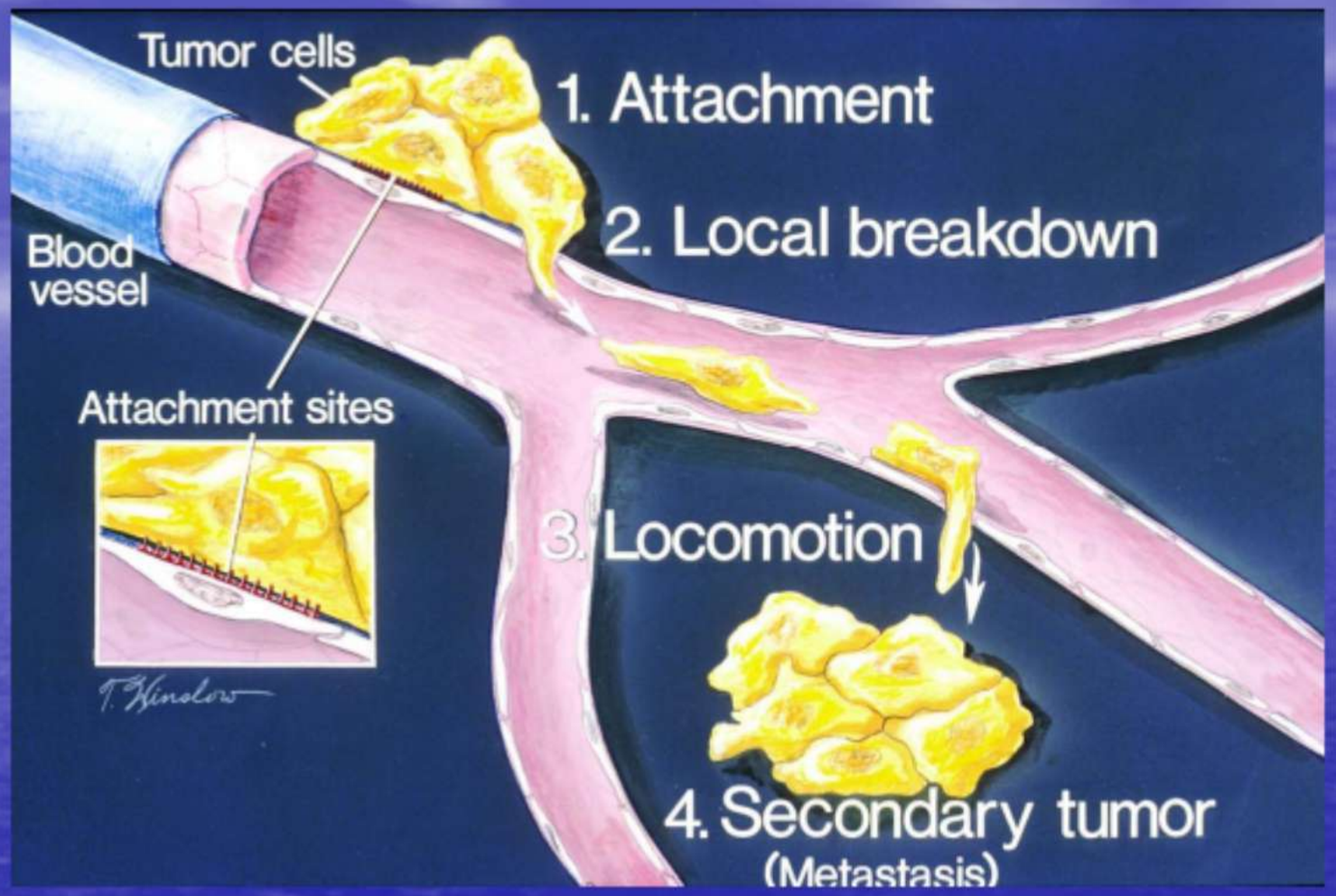


A variety of target genes have been identified that play roles in the various cellular responses to *p53*. Each of these genes has been shown to contain *p53* binding sites within its regulatory regions.

From Benign to Malignant

- Cancer cells divide too quickly and can leave the original site and enter the blood, lymph, or tissues.
- Most cells divide a set number (60-70) of times, then they stop dividing.
- This usually limits benign tumors to small sizes.
- Cancer cells can divide indefinitely.





The vessels of the circulatory and lymphatic systems provide a pipeline for cancer cells to move to other locations in the body through a process called *metastasis*.

Growth characteristics of cells in culture

(parallels what happens in vivo)

- Normal primary cells: survive only a relatively short time in culture (days to weeks)
- Immortalized (established) cells: immortalized, but not transformed (they retain many of their in vivo characteristics)
  - anchorage dependence (prefer to adhere to culture plate)
  - serum dependence (require growth factors)
  - contact inhibited (stop growing when cells are confluent)
- Transformed cells: will form tumors (tumorigenic) when injected into a host, but will not necessarily kill the host
  - less dependent on substratum
  - less dependent on growth factors
  - not contact inhibited
- Metastatic cells: fully transformed cells that have the ability to migrate and invade tissues; will establish new colonies and kill the host

THANK YOU